

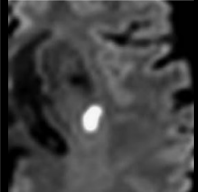
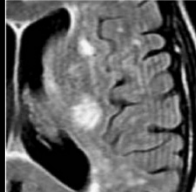
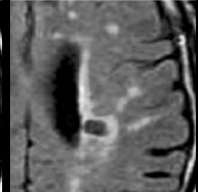
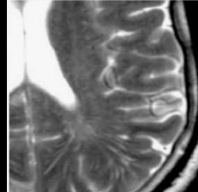
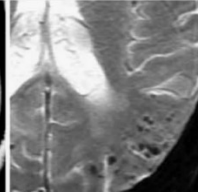
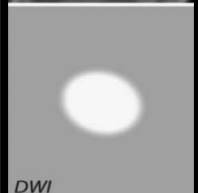
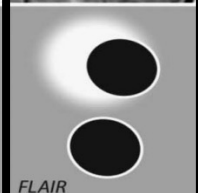
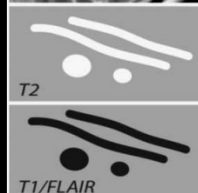
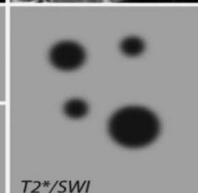
Lacunar Stroke

Outline

1. Definition
2. Pathology
3. Epidemiology
4. Clinical Syndromes
5. Diagnosis
6. Management Considerations
7. Prognosis

Cerebral Small Vessel Disease

- Various pathologies affecting small arteries, arterioles, capillaries, and venules
 - Arteriolosclerosis
 - Genetic, inflammatory / immune, other
- Small cortical arteries/arterioles + penetrating arteries

	Recent small subcortical infarct	White matter hyperintensity	Lacune	Perivascular space	Cerebral microbleeds
Example image					
Schematic					
	DWI	FLAIR	FLAIR	T2 T1/FLAIR	T2*/SWI
Usual diameter ¹	≤ 20 mm	variable	3-15 mm	≤ 2 mm	≤ 10 mm
Comment	best identified on DWI	located in white matter	usually have hyperintense rim	usually linear without hyperintense rim	detected on GRE seq., round or ovoid, blooming
DWI	↑	↔	↔/(↓)	↔	↔
FLAIR	↑	↑	↓	↓	↔
T2	↑	↑	↑	↑	↔
T1	↓	↔/(↓)	↓	↓	↔
T2* / GRE	↔	↑	↔ (↓ if haemorrhage)	↔	↓↓

History

- 1838: Lacune → lacuna (Latin: cavity)
 - *état criblé* → enlarged spaces in WM
 - 1901: Pierre Marie & Ferrand → capsular, lenticular, and pontine infarction
 - Mid-20th century: C. Miller Fisher → clinicopathologic characterization
 - Clinical syndromes
 - Lipohyalinosis
- 
- A black and white portrait of a middle-aged man with receding hair, wearing glasses and a dark suit jacket over a light-colored shirt. He is looking slightly to the right of the camera with a neutral expression.



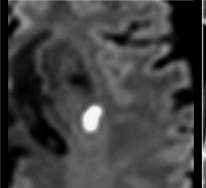
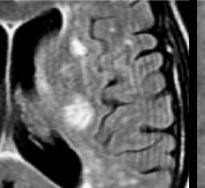
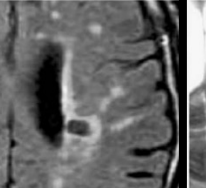
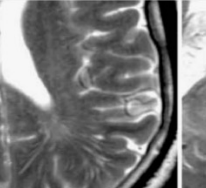
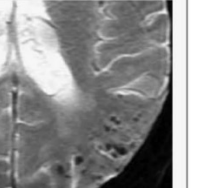
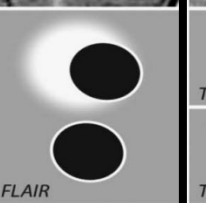
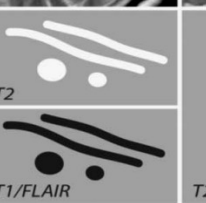
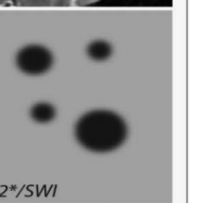
Neurology Jan 1986, 36 (1) 85.

J Stroke Cerebrovasc Dis. 2019 Aug;28(8):2079-2097.

Neurology. 1965 Aug;15:774-84.

Evolution of the Definition

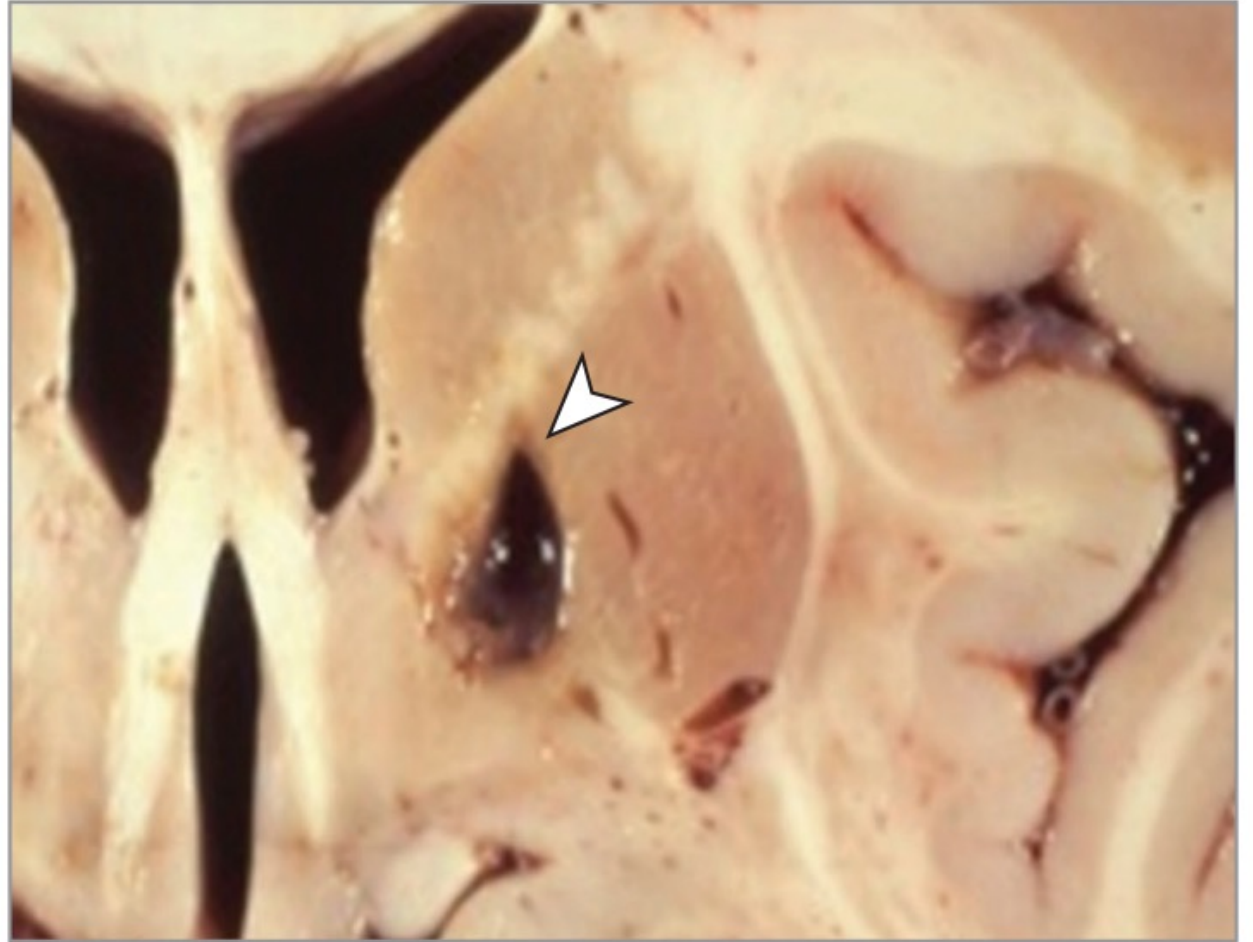
- Lacune (neuropathologic term: cavitation from small deep brain infarct) → lacunar infarct (precavitary phase)
- Lacunar stroke (clinical entity) & lacunar syndrome (neurologic features)
- **ST**andards for **R**eporting **V**ascular changes on **n**euroimaging (STRIVE)

	Recent small subcortical infarct	White matter hyperintensity	Lacune	Perivascular space	Cerebral microbleeds
Example image					
Schematic					
	DWI	FLAIR	FLAIR	T1/FLAIR	T2*/SWI
Usual diameter ¹	≤ 20 mm	variable	3-15 mm	≤ 2 mm	≤ 10 mm
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DWI	↑	↔	↔/(↓)	↔	↔
FLAIR	↑	↑	↓	↓	↔
T2	↑	↑	↑	↑	↔
T1	↓	↔/(↓)	↓	↓	↔
T2* / GRE	↔	↑	↔ (↓ if haemorrhage)	↔	↓↓

Location

- Basal ganglia (putamen), thalamus, WM – IC, pons
 - WM of cerebral gyri
- Lenticulostriate, thalamoperforating, paramedian branches
 - Small size (diameter 40 - 900 μm)
 - Proximal origin
 - Lack of collateral circulation

B Cavity in medial basal ganglia



Vascular Pathology

- Microangiopathy – arterial media thickening
 - Arteriolosclerosis (40-150 μm)
 - Lipohyalinosis / fibrinoid necrosis (40-200 μm)
- Branch atheromatous disease (200-800 μm)
 - Parent large artery
 - Ostium or proximal segment of penetrator

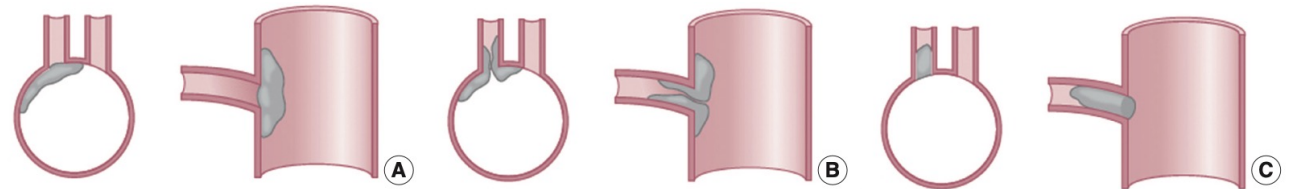
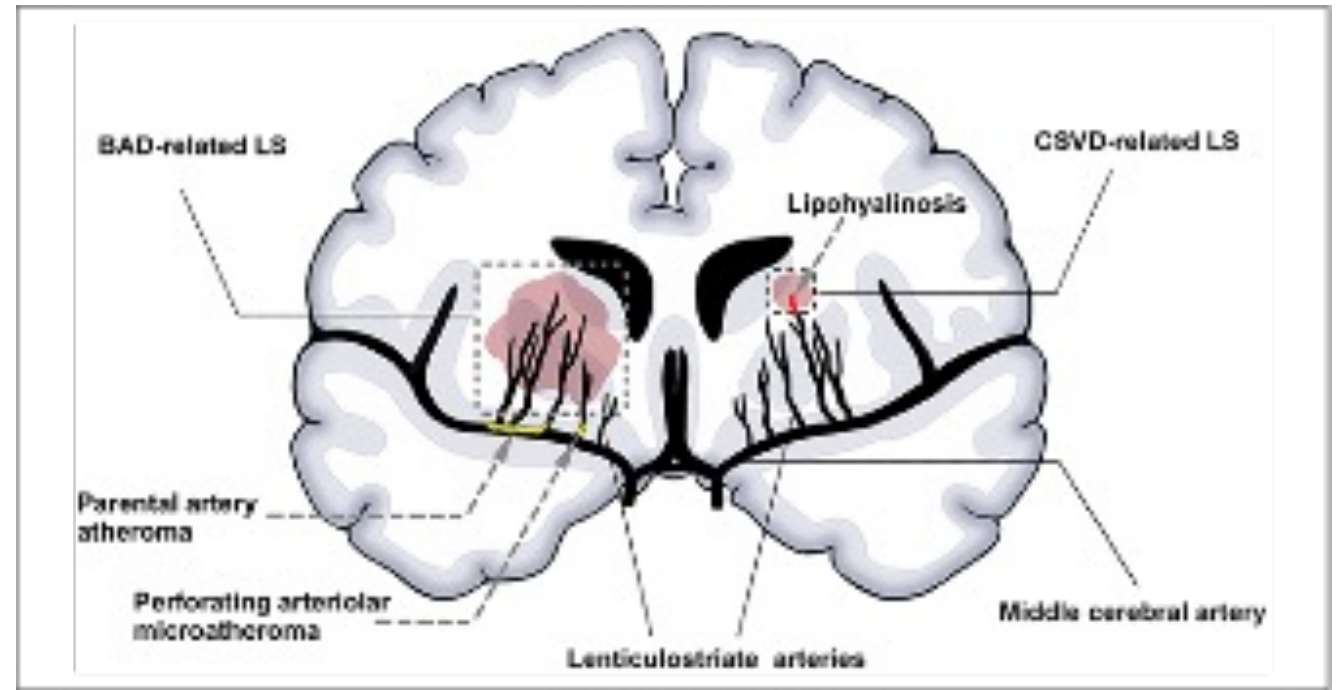
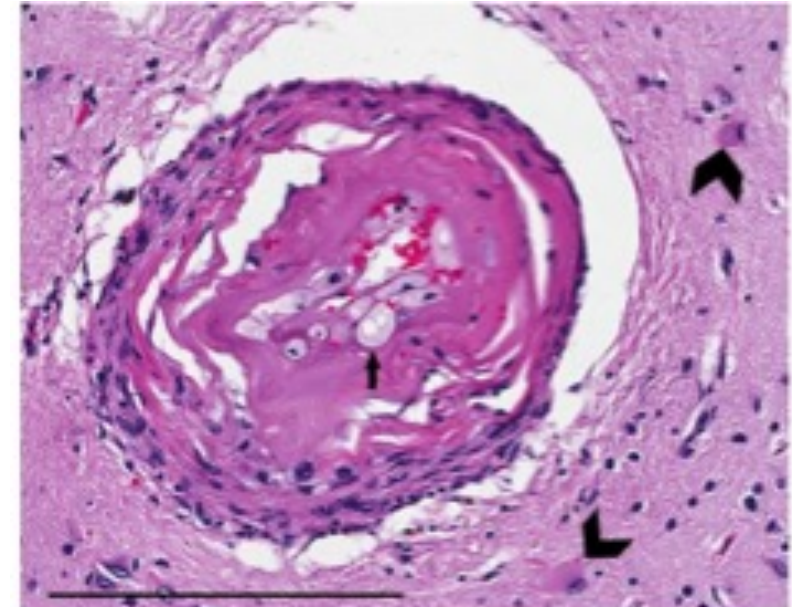
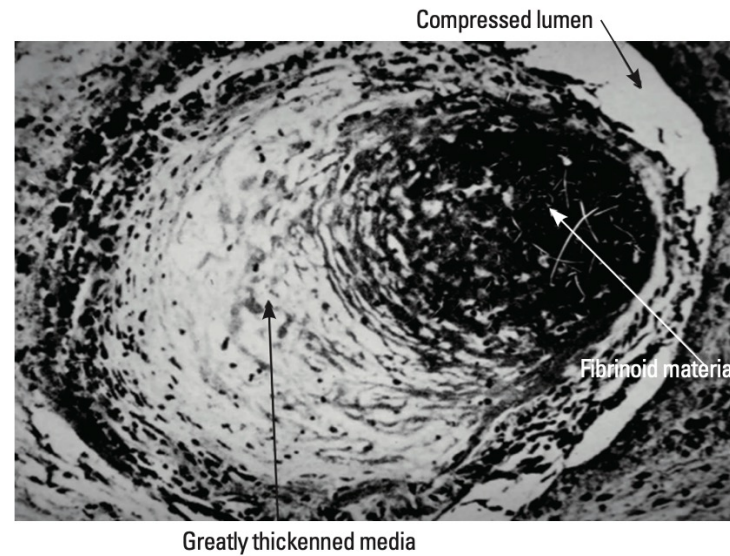
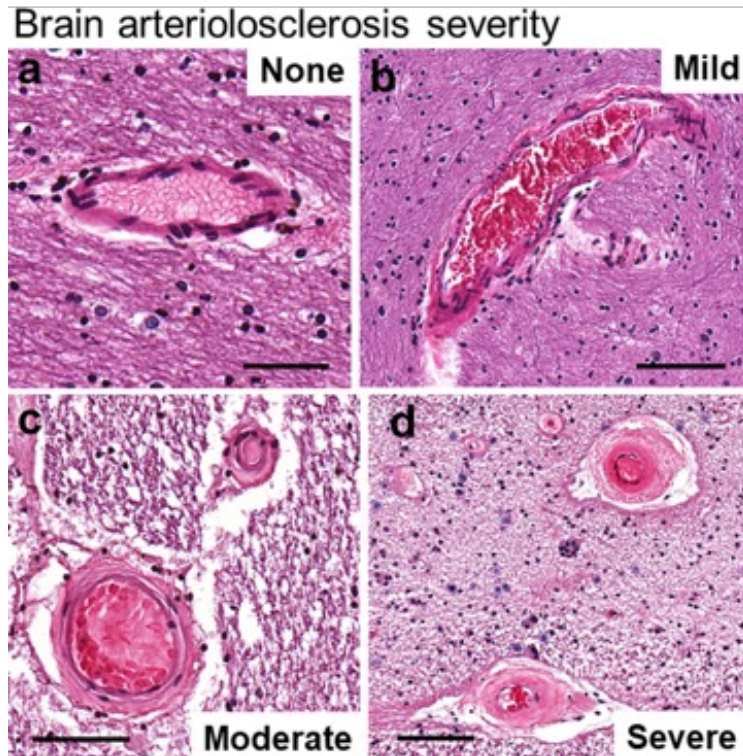


Figure 3. Drawing showing the arterial pathology in atheromatous branch disease: (A) plaque in parent artery obstructing a branch, (B) junctional plaque extending into the branch, (C) microatheroma formed at the orifice of a branch.

Arteriolosclerosis & Lipohyalinosis



Branch Atheromatous Disease

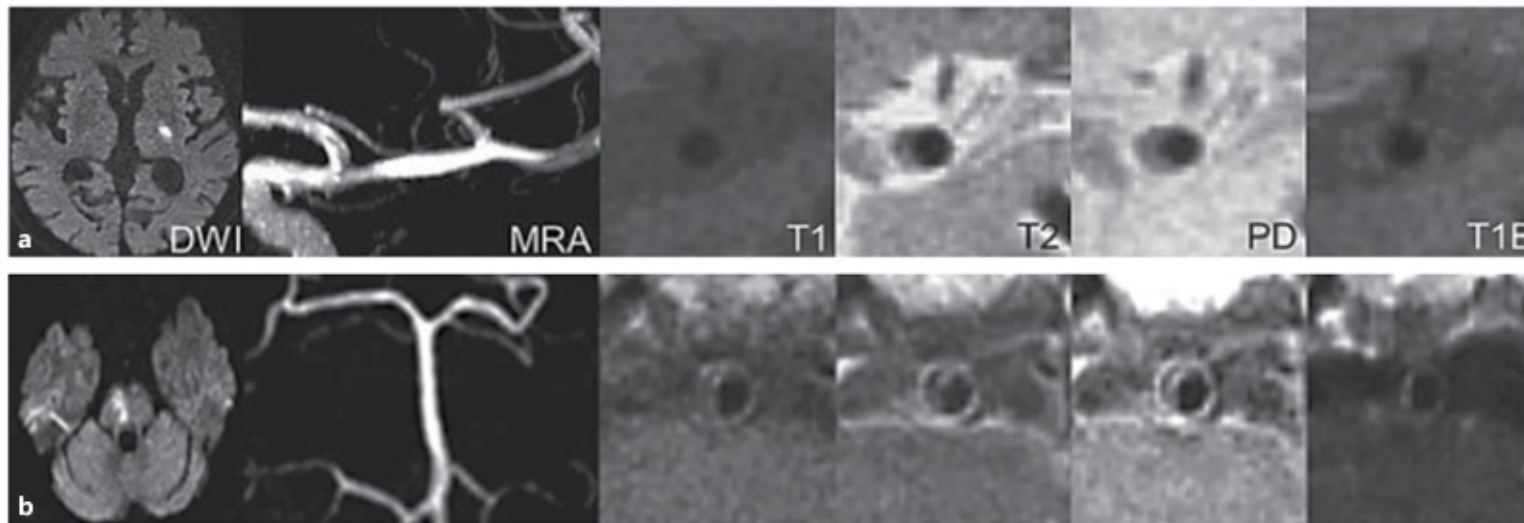
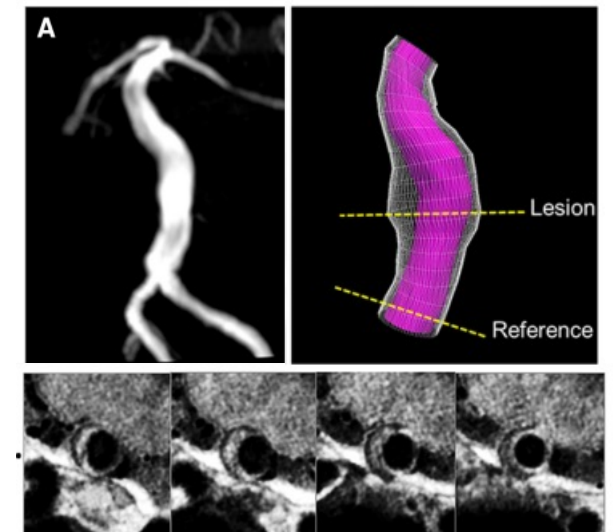
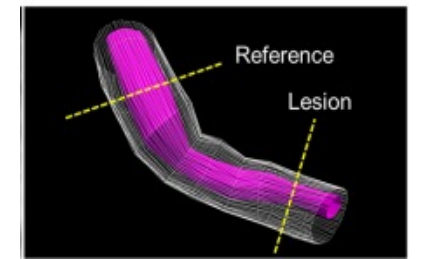
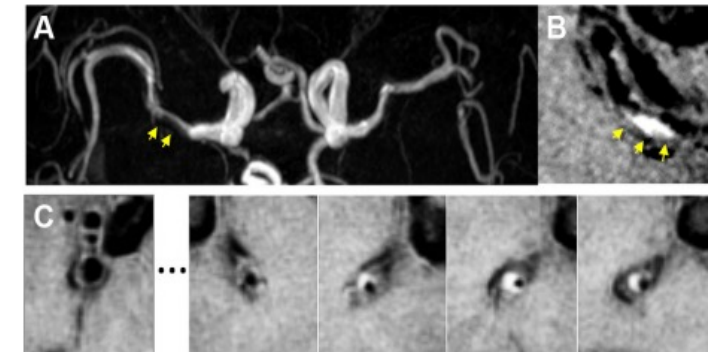


Fig. 3. HR MRI findings of BAD-related infarct and branch atheromatous plaque in the middle cerebral artery (a) and in the basilar artery (b).



Embolism

- Embolism
 - Causal relationship unclear
- Carotid stenosis & cardiac sources of embolism
 - Lacunar infarction < cortical infarction

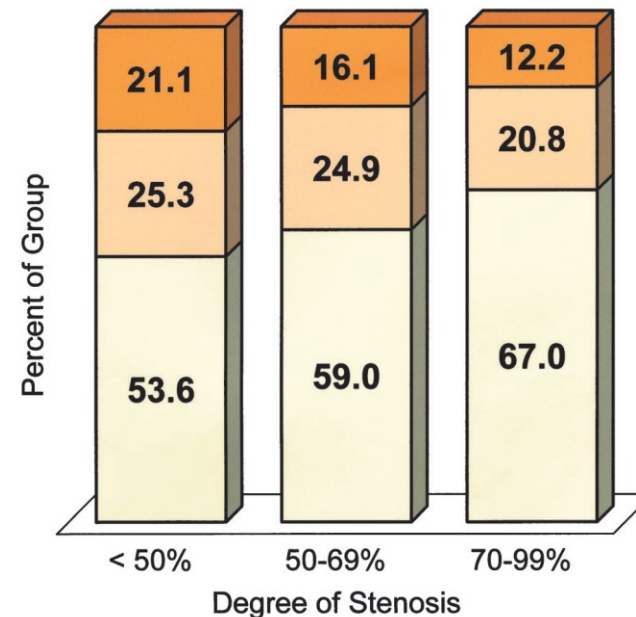


Figure 2. Relationship between category of stroke and degree of internal carotid artery (ICA) stenosis. Lacunar stroke was more likely to occur in patients with milder degrees of ICA stenosis than with severe (70 to 99%) stenosis. Probable lacunar (top)—possible lacunar (middle)—non-lacunar (bottom).

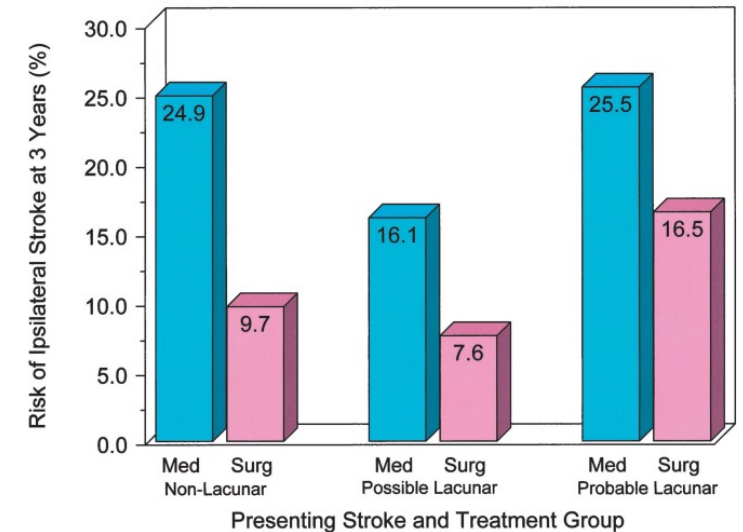


Figure 4. Risk of ipsilateral stroke at 3 years by presenting stroke category and treatment group for patients with 50 to 99% internal carotid artery stenosis. The numbers of patients represented in each bar, from left to right, are 172, 160, 57, 69, 41, and 38. Med = medically treated; Surg = surgically treated.

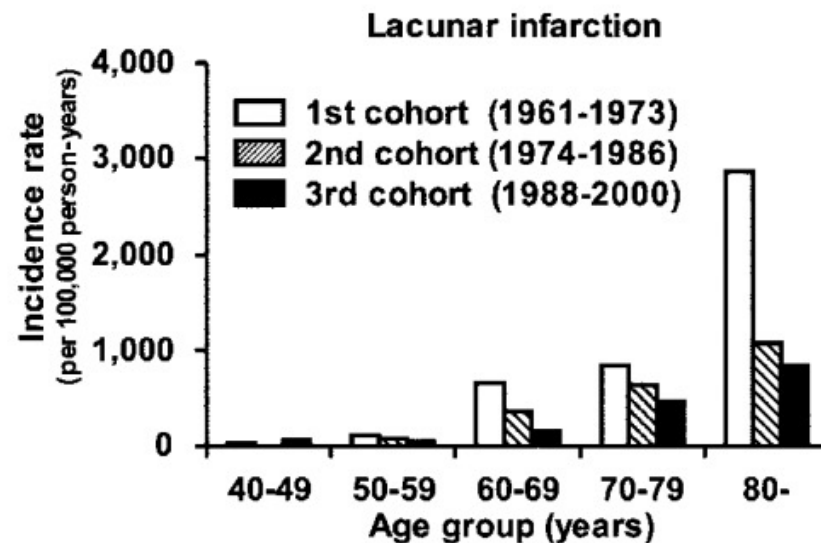
Stroke. 1991 Oct;22(10):1236-41.

Stroke. 1991 Feb;22(2):175-81.

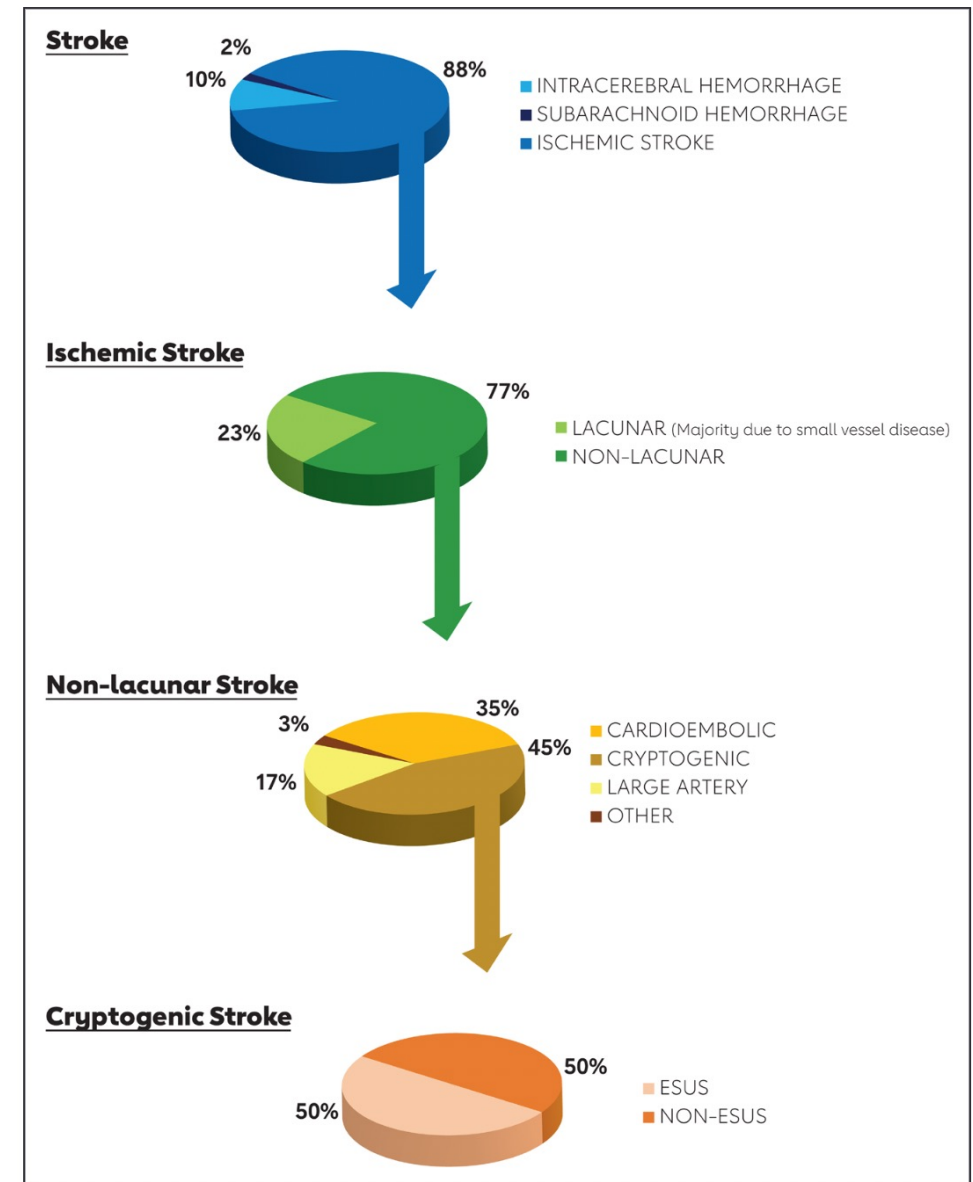
Neurology. 2000 Feb 8;54(3):660-6.

Epidemiology

- ~ ¼ of ischemic stroke
- GCNKSS: incidence ↑ in Black Americans



Stroke. 2021 Jul;52(7):e364-e467.
Stroke. 1999 Dec;30(12):2517-22.
Neurology. 2006 May 23;66(10):1539-44.



Source-qualified teaching update | 30 September 2026

The stroke-subtype proportions reproduced from AHA/ASA 2021 Figure 1 are approximate and vary with population and diagnostic testing.

doi.org/10.1161/STR.0000000000000375

Risk Factors

- Shared with other IS subtypes
- Age, Hypertension, Diabetes, Tobacco use
 - LDL-C, hyperhomocystinemia

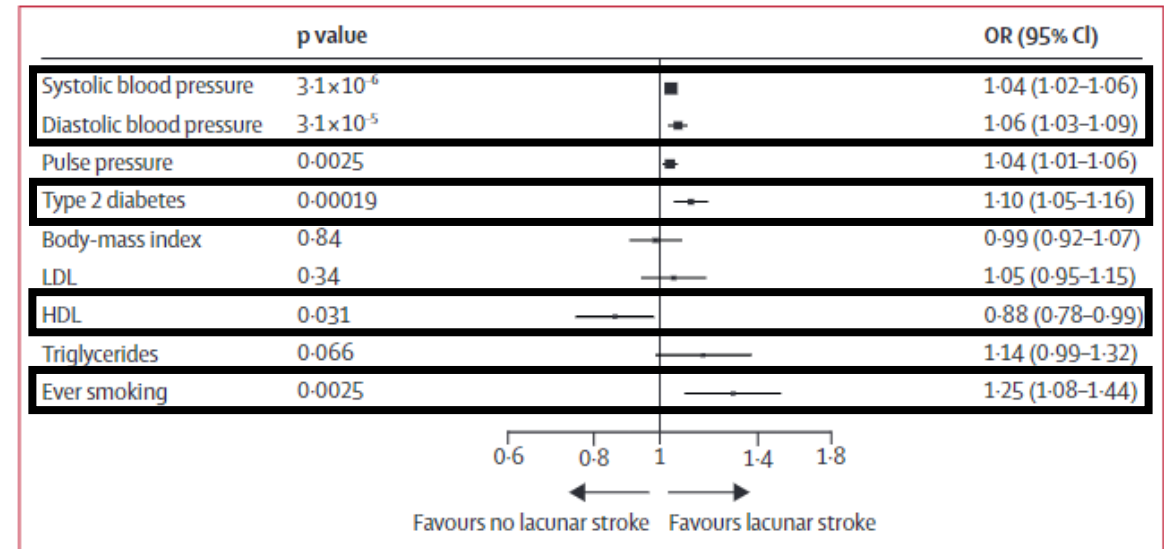


Figure 4: Associations between genetically proxied cardiovascular risk factors and lacunar stroke from mendelian randomisation analysis using the inverse variance weighted method
 Estimates are presented as ORs per genetically proxied increase in each risk factor (original scale). OR=odds ratio.

Stroke. 2005 Apr;36(4):891-901.
 Lancet Neurol. 2021 May;20(5):351-361.

Source-qualified teaching update | 30 September 2026

Distinguish observed associations from causality. The displayed HDL genetic association (nominal $P=.031$) did not meet the study's multiple-testing threshold; LDL was not associated in that analysis. This does not establish that LDL-lowering treatment lacks clinical benefit.

pmc.ncbi.nlm.nih.gov/articles/PMC8062914/
pmc.ncbi.nlm.nih.gov/articles/PMC2577185/

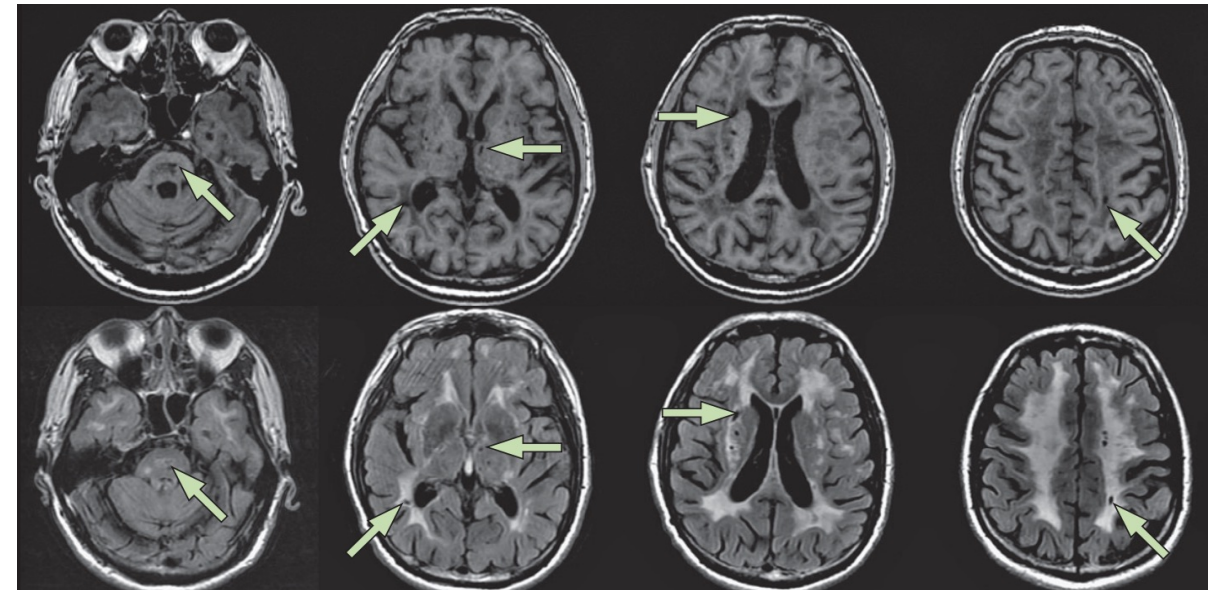
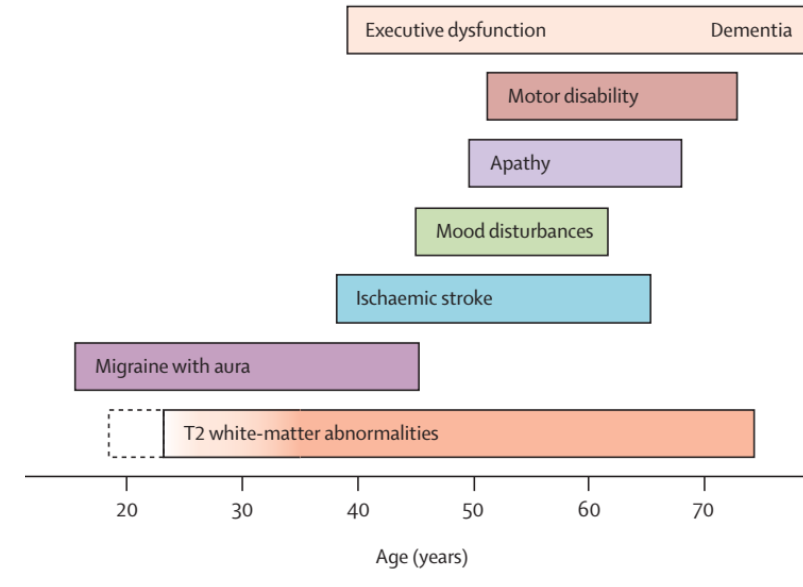
Genetics

	Chromosome	Base position*	Genomic context	Identifier	RA/OA	RAF	Lacunar stroke (European: 6030 cases, 219 389 controls)			Lacunar stroke (transethnic: 7338 cases, 225 258 controls)	
							OR (SE)	p value	Studies	OR (SE)	p value
Genome-wide significance for lacunar stroke											
ICA1L-WDR12-CARF-NBEAL1†	2	203 968 973	Intronic	rs72934535	T/C	0.89	1.25 (0.04)	3.7 × 10 ⁻⁹	12	1.22 (0.04)	5.2 × 10 ⁻⁸
ULK4†	3	41839 370	Intronic	rs4621303	T/A	0.83	1.15 (0.03)	1.7 × 10 ⁻⁷	14	1.16 (0.03)	6.4 × 10 ⁻⁹
SPI1-SLC39A13-PSMC3-RAPSN	11	47 434 986	Exonic	rs2293576	G/A	0.67	1.14 (0.02)	7.2 × 10 ⁻¹⁰	14	1.14 (0.02)	6.0 × 10 ⁻¹⁰
ZCCHC14	16	87 575 332	Intergenic	rs12445022	A/G	0.34	1.13 (0.02)	2.5 × 10 ⁻⁸	13	1.12 (0.02)	9.0 × 10 ⁻⁸
ZBTB14-EPB41L3	18	5 389 832	Intergenic	rs9958650	G/A	0.10	1.18 (0.03)	9.9 × 10 ⁻⁷	12	1.19 (0.03)	2.4 × 10 ⁻⁸
Genome-wide significance in multi-trait analysis											
SLC25A44-PMF1-BGLAP†	1	156 197 380	Intronic	rs2984613	C/T	0.64	1.10 (0.02)	2.5 × 10 ⁻⁵	13	1.09 (0.02)	1.4 × 10 ⁻⁵
LOX-ZNF474-LOC100505841	5	121 518 378	Downstream	rs2303655	T/C	0.81	1.14 (0.03)	3.6 × 10 ⁻⁵	11	1.12 (0.03)	0.00014
FOXF2-FOXQ1	6	1 366 718	Intergenic	rs7766042	C/T	0.11	1.17 (0.03)	3.7 × 10 ⁻⁶	11	1.18 (0.03)	1.2 × 10 ⁻⁶
VTA1-GPR126	6	142 562 417	Intergenic	rs225744	C/T	0.77	1.11 (0.03)	3.5 × 10 ⁻⁵	12	1.09 (0.02)	0.00050
SH3PXD2A	10	105 447 838	Intronic	rs61000833	T/C	0.60	1.10 (0.02)	1.7 × 10 ⁻⁵	12	1.07 (0.02)	0.0024
HTRA1-ARMS2	10	124 233 181	Intronic	rs79043147	T/C	0.07	1.21 (0.04)	3.2 × 10 ⁻⁶	11	1.22 (0.04)	1.1 × 10 ⁻⁶
COL4A2	13	111 040 681	Intronic	rs11838776	A/G	0.29	1.11 (0.02)	4.3 × 10 ⁻⁶	12	1.11 (0.02)	1.6 × 10 ⁻⁶

- ECM disruption, pericyte differentiation, TGF- β signaling, myelination
- Overlap with WMH
- Monogenic small vessel stroke
 - HTRA1
 - COL4A2

Monogeneic Disorders

- CADASIL (cerebral AD arteriopathy w/ subcortical infarcts & leukoencephalopathy)
 - NOTCH3
- CARASIL
 - HTRA1
 - alopecia, spondylosis
- RVCL-S (retinal vasculopathy w/ cerebral leukodystrophy & systemic manifestations)
 - TREX1
 - Raynaud's, cirrhosis, renal dysfunction, osteonecrosis



Clinical Syndromes

- Pure motor hemiparesis
 - Pure sensory syndrome
 - Sensorimotor syndrome
 - Dysarthria-clumsy hand syndrome
 - Ataxic hemiparesis
-
- Lack cortical findings
- * Moderate-High PPV for radiological lacunar infarct

Modified pure motor hemiparesis with motor aphasia

Pure motor hemiparesis sparing face

Mesencephalo-thalamic syndrome

Thalamic dementia

Pure motor hemiparesis with horizontal gaze palsy

Pure motor hemiparesis with crossed third-nerve palsy (Weber syndrome)

Pure motor hemiparesis with crossed sixth-nerve palsy

Pure motor hemiparesis with confusion

Cerebellar ataxia with crossed third-nerve palsy (Claude syndrome)

Hemiballismus

Lower basilar branch syndrome -- dizziness, diplopia, gaze palsy, dysarthria, cerebellar ataxia, trigeminal numbness

Lateral medullary syndrome

Lateral pontomedullary syndrome

Locked-in syndrome (bilateral pure motor hemiparesis)

Pure dysarthria

Acute dystonia of thalamic origin

Lacunar state

Neurology. 1982 Aug;32(8):871-6.

Neurology. 1997 May;48(5):1204-11.

Stroke. 2004 Jun;35(6):1430-5.

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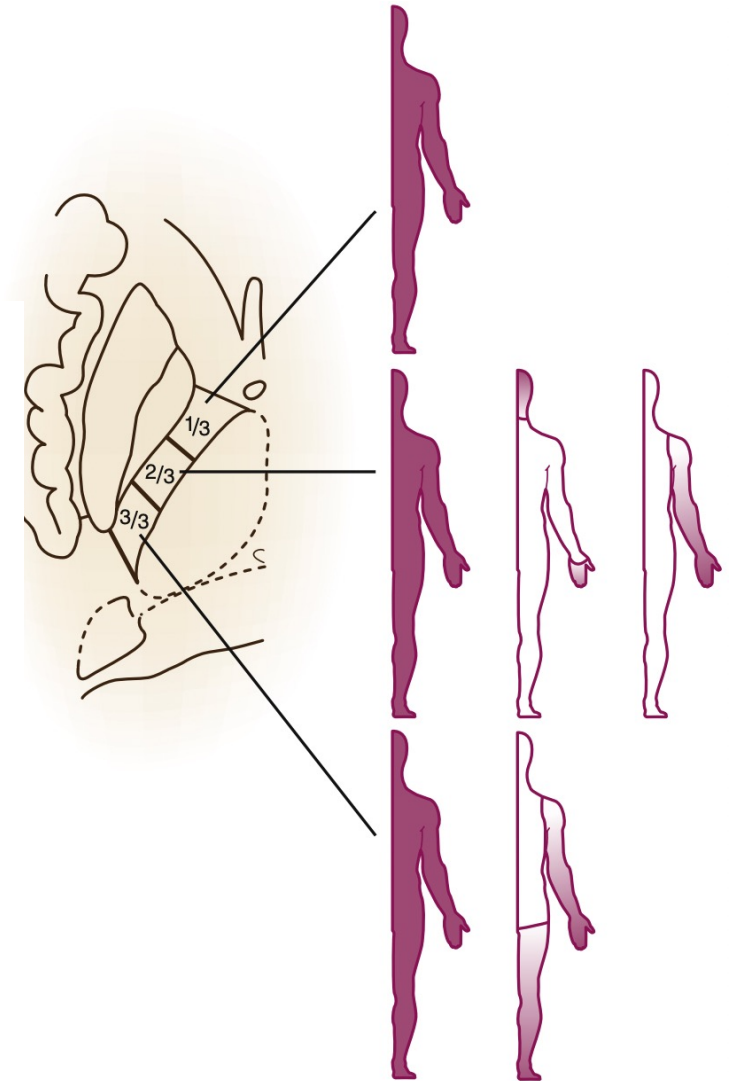
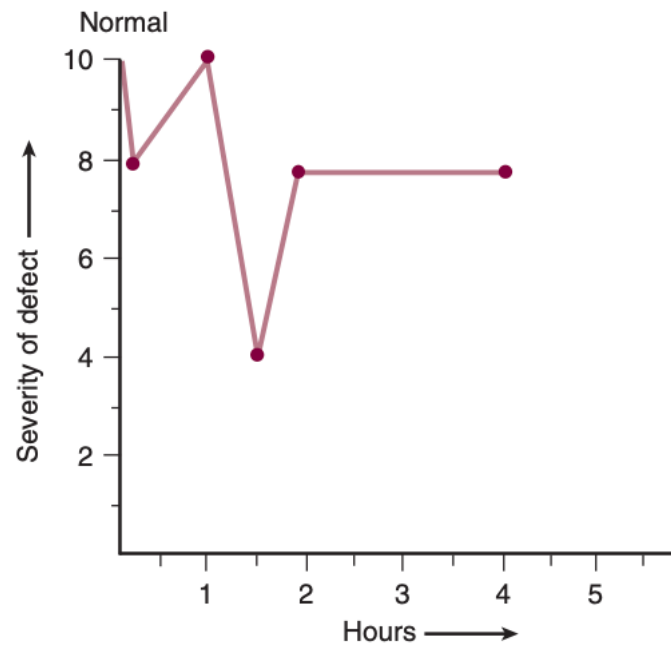
The right-hand list combines heterogeneous reported presentations and lesion locations. These are not each validated markers of lacunar small-vessel disease; a clinical syndrome alone does not establish its mechanism.

pubmed.ncbi.nlm.nih.gov/7048128/

pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

Pure Motor Hemiparesis

- Face / Arm / Leg
- Internal capsule, pons > corona radiata, medullary pyramid
- PPV: 52-85%



Stroke: Pathophysiology, Diagnosis, and Management, 7th Edition
Caplan's Stroke, 5th Edition
Stroke. 1991;22:175–181.

Source-qualified teaching update | 30 September 2026

Capsular warning syndrome describes recurrent, stereotyped crescendo episodes; a single preceding TIA is not sufficient to establish it.

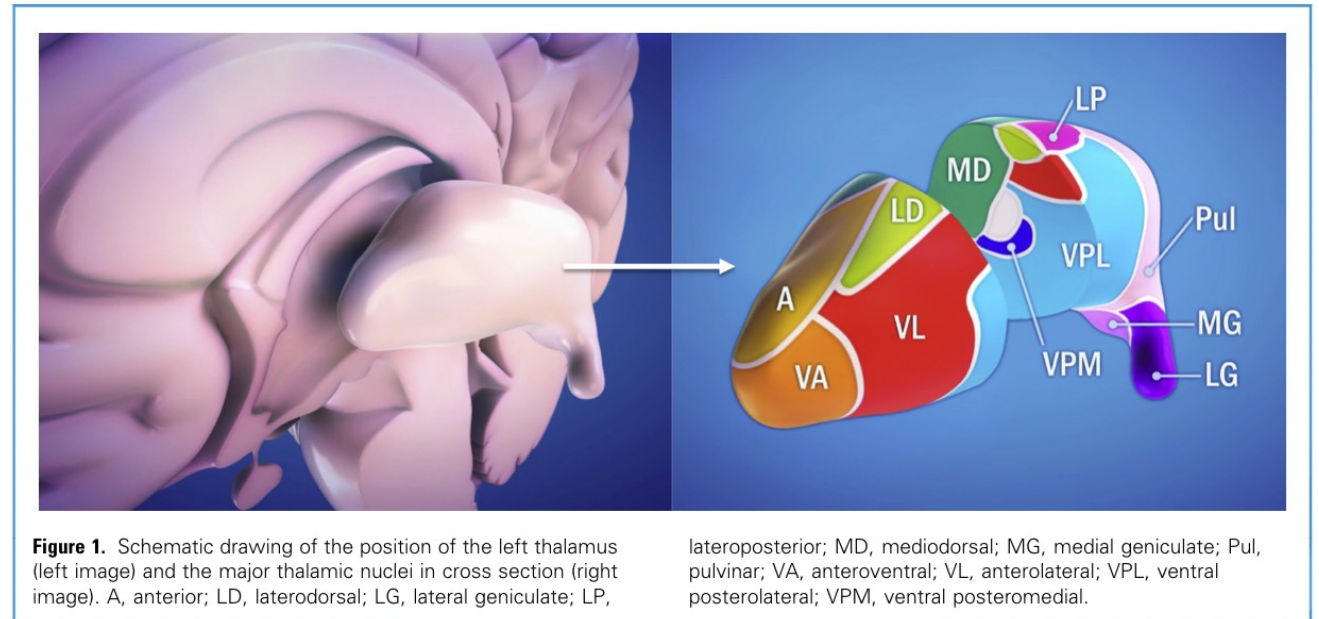
pubmed.ncbi.nlm.nih.gov/8492952/

pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

Pure Sensory Syndrome

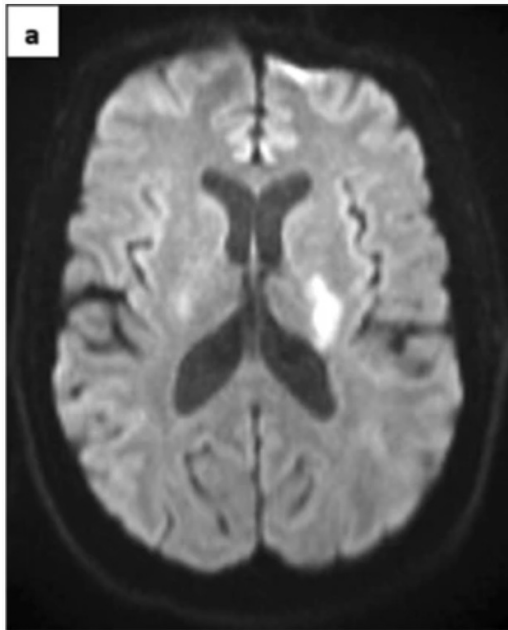
- Face / Arm / Leg
- Thalamus > pons, corona radiata
- PPV: 95-100%

- * Midline extension
- * Cheiro-oral, cheiro-oral-pedal syndrome
- * Central post-stroke pain



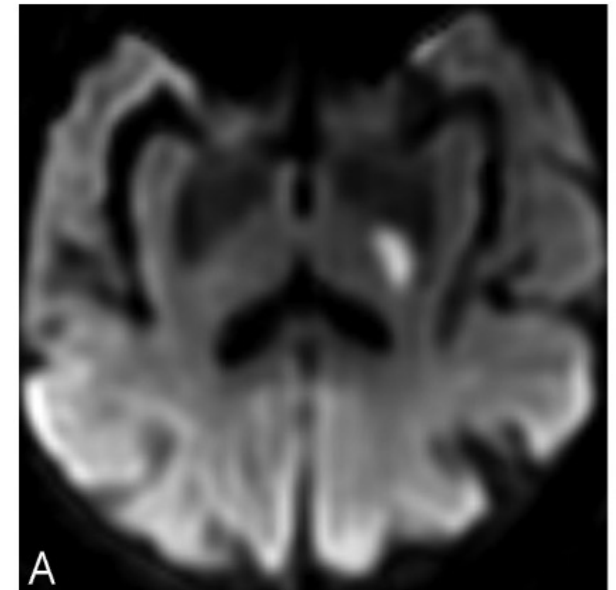
Sensorimotor Syndrome

- Face / Arm / Leg
- Thalamocapsular > pons, medulla
- PPV: 51-87%



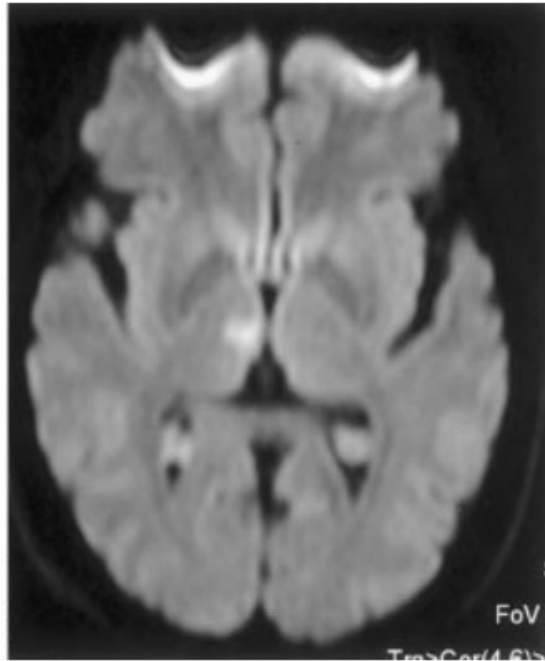
Dysarthria-Clumsy Hand Syndrome

- Dysarthria, Hand Ataxia / Paresis + Ipsilateral Facial Paresis, Dysphagia
- Internal capsule > basal pons, corona radiata
- PPV: 96%



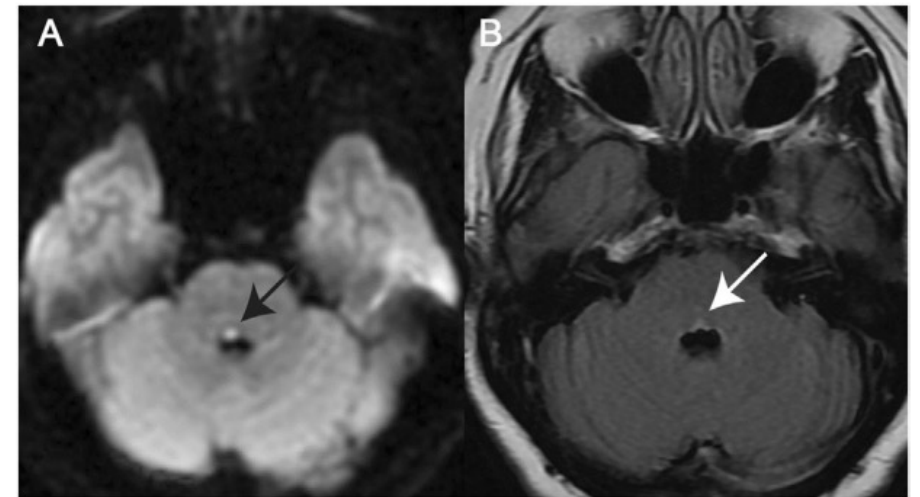
Ataxic Hemiparesis

- Ipsilateral limb ataxia > paresis
- Internal capsule, pons > corona radiata, thalamus
- PPV: 59-95%



Other Syndromes

- Brainstem (INO, motor syndrome + CN palsy)
- Movement disorder (hemichorea-hemiballismus, dystonia)
- Vascular cognitive impairment & dementia



Front Neurol. 2017 Sep 22;8:502.
Neurology. 2003 Aug 12;61(3):334-8.

Source-qualified teaching update | 30 September 2026

Original INO illustration: panel A (patient 7) is ischemic infarction; panel B (patient 19) is demyelination. The images illustrate the differential and must not both be labeled lacunar infarction.

pmc.ncbi.nlm.nih.gov/articles/PMC5614936/

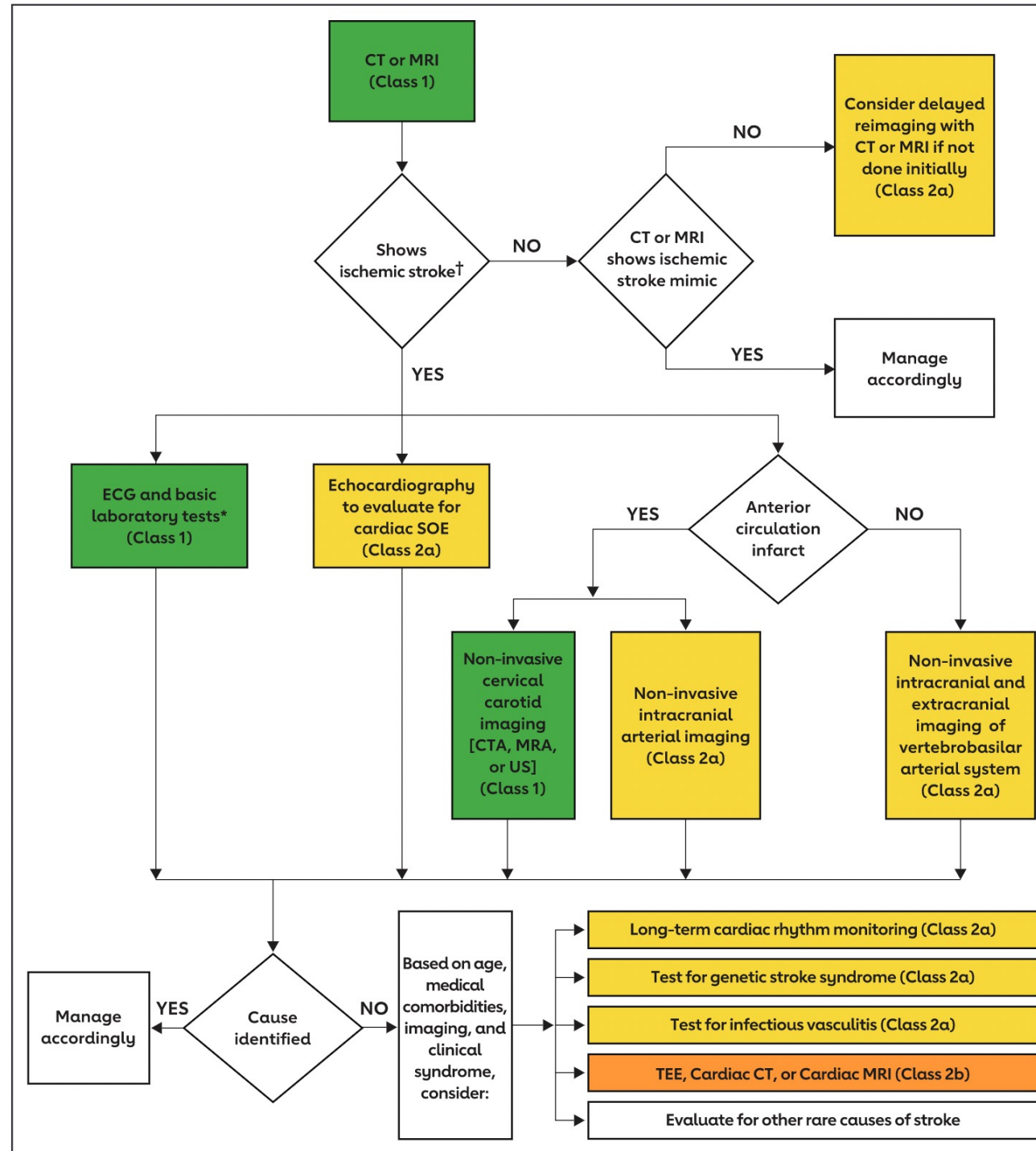
Evaluation

CENTRAL ILLUSTRATION Algorithm for Triaging Patients With Acute Ischemic Stroke With Different Imaging Options

Acute-treatment algorithm retired.

Use current treatment-specific eligibility guidance. See the source-qualified update below.

J Am Coll Cardiol. 2020 Apr 21;75(15):1844-1856.
Stroke. 2021 Jul;52(7):e364-e467.



Source-qualified teaching update | 30 September 2026

The incomplete 2020 acute-treatment algorithm has been retired. Current reperfusion decisions require treatment-specific time, clinical and imaging criteria and appropriate stroke-unit care. The retained AHA 2021 diagram concerns secondary-prevention evaluation: a clinical TIA merits comparable etiologic evaluation even without a visible infarct, with appropriate basic laboratory testing.

pubmed.ncbi.nlm.nih.gov/32299596/

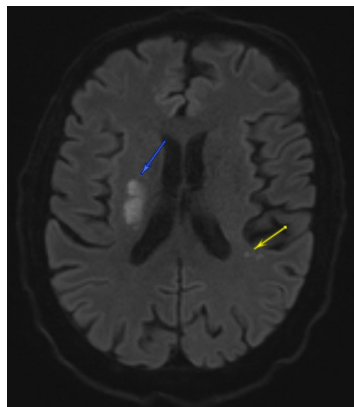
doi.org/10.1161/STR.0000000000000513

doi.org/10.1161/STR.0000000000000375

Imaging

CT

- Difficult to visualize lacunar infarcts <2mm
*brainstem
- Sensitivity → 30-44%



MRI - DWI

- High sensitivity & specificity
- Acute: ≤20 mm
- Chronic: <15 mm

	Recent small subcortical infarct	Lacune	Perivascular space
Example image			
Schematic			
	DWI	FLAIR	T2
			T1/FLAIR
Usual diameter ¹	≤ 20 mm	3-15 mm	≤ 2 mm
Comment	best identified on DWI	usually have hyperintense rim	usually linear without hyperintense rim
DWI	↑	↔/(↓)	↔
FLAIR	↑	↓	↓
T2	↑	↑	↑
T1	↓	↓	↓
T2* / GRE	↔	↔ (↓ if haemorrhage)	↔

Lancet Neurol. 2013 Aug;12(8):822-38.

Source-qualified teaching update | 30 September 2026

Multiple acute lesions increase suspicion for an alternative source; they do not prove embolism. STRIVE dimensions are approximate axial conventions, not stand-alone diagnostic cutoffs. A lacune is a cavity with additional imaging features; clinical small-vessel events may lack a visible DWI lesion.

pubmed.ncbi.nlm.nih.gov/15746460/

pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

pmc.ncbi.nlm.nih.gov/articles/PMC3714437/

Variable Imaging Evolution

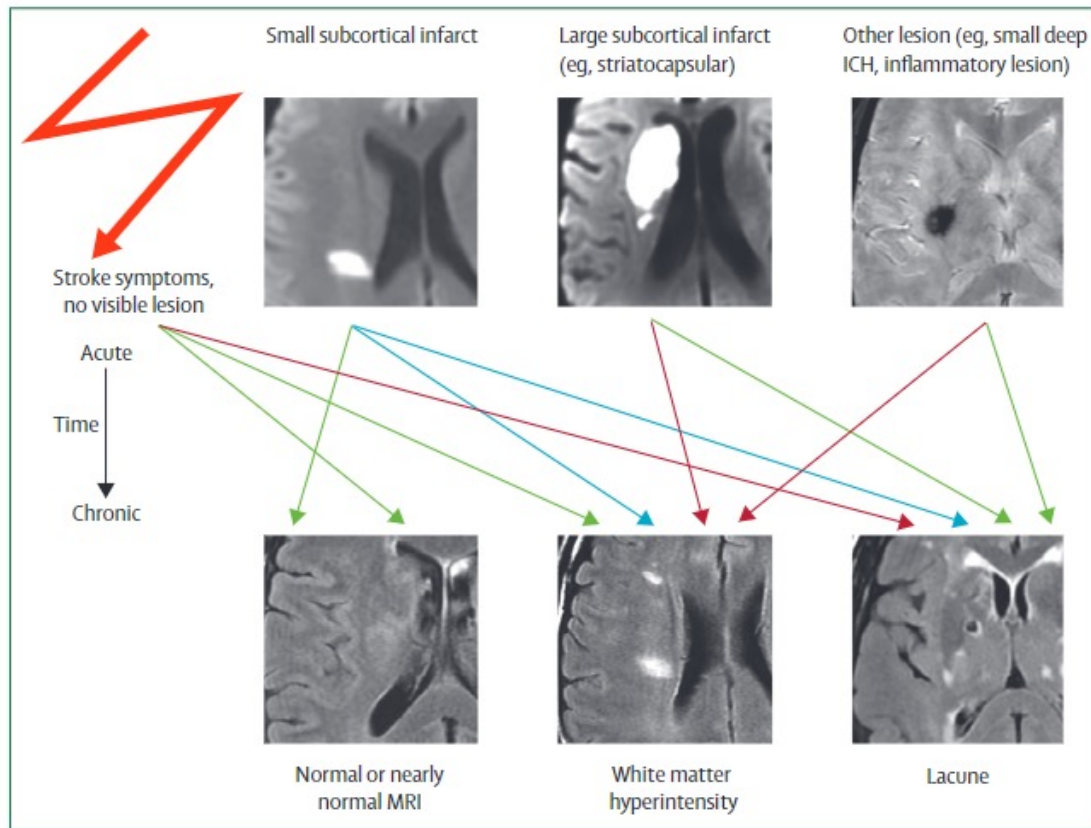
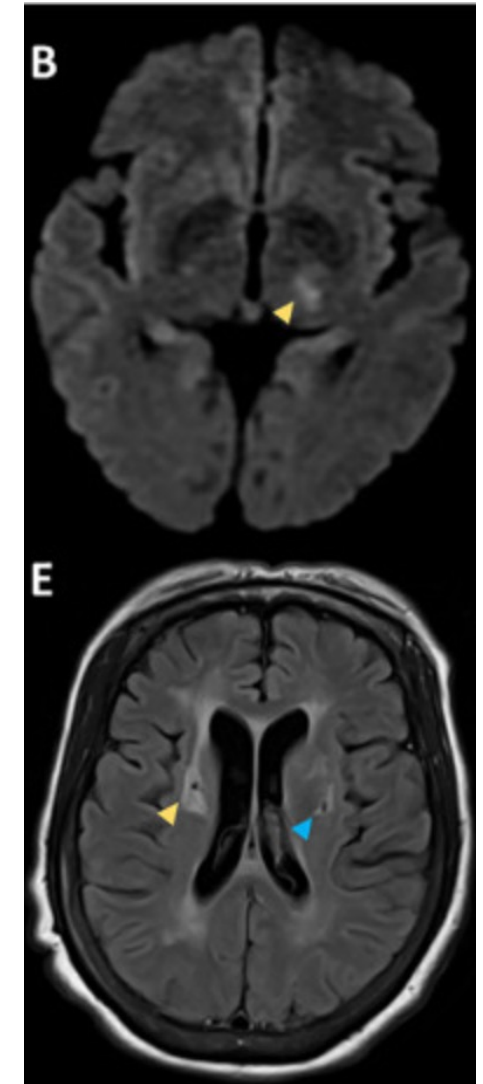
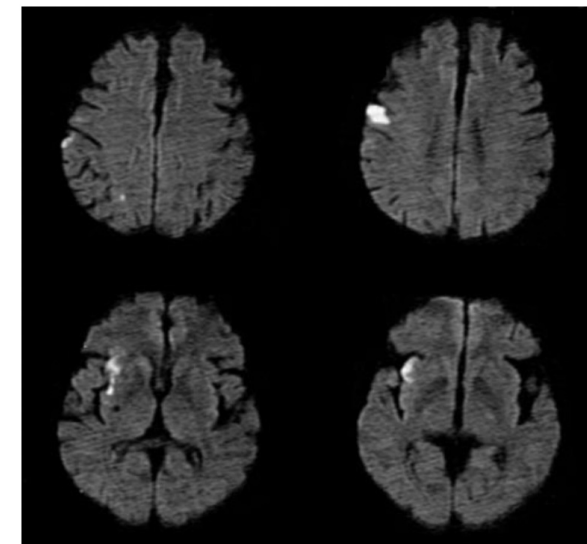
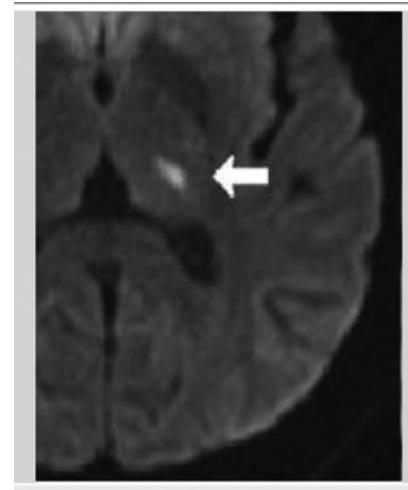


Figure 1: Variable fates of lesions related to small vessel disease and the convergence of acute lesions with different causes but similar late appearances on MRI

1. Cavitated, surrounding gliosis
 2. Subcortical/periventricular WMH
 3. No evidence of prior infarct
- Subcortical cavitated lesion → T2*-WI to eval for hemorrhage

Diagnosis

- Lacunar syndrome
- CT, vascular imaging, CTP not sensitive
- MRI
 - Presentation $\neq$ Stroke Subtype



Stroke. 2005 Apr;36(4):757-61.

Source-qualified teaching update | 30 September 2026

Diagnosis integrates a compatible clinical event, imaging and exclusion of plausible alternatives. Risk factors support the assessment but are not obligatory; a visible single-perforator lesion is not required in every clinical lacunar event.

pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

pmc.ncbi.nlm.nih.gov/articles/PMC3714437/

Management

IV-tPA

Capsular Warning Syndrome / Infarct Progression

Table 1. Association of Baseline Covariates With Outcome (OR Based on Multiple Outcomes⁵: Univariate Analyses and Treatment Interactions)

Variable	OR	95% CI	P ¹	Treatment Interaction P ²
Baseline stroke subtypes ³			.009	.78
Cardioembolic vs small	0.44	0.28-0.68		
Large vs small	0.45	0.29-0.71		
Other vs small	0.69	0.27-1.71		

- Extension of thrombosis
- Progressive branch occlusion
- Hemodynamic failure
- Peri-infarct edema

Stroke. 1997 Nov;28(11):2119-25
Int J Stroke. 2012 Jun;7(4):321-9..

Source-qualified teaching update | 30 September 2026

Thrombus extension, branch occlusion, hemodynamic failure and edema are proposed mechanisms of progression. The list does not establish a mechanism-directed treatment with proven benefit.

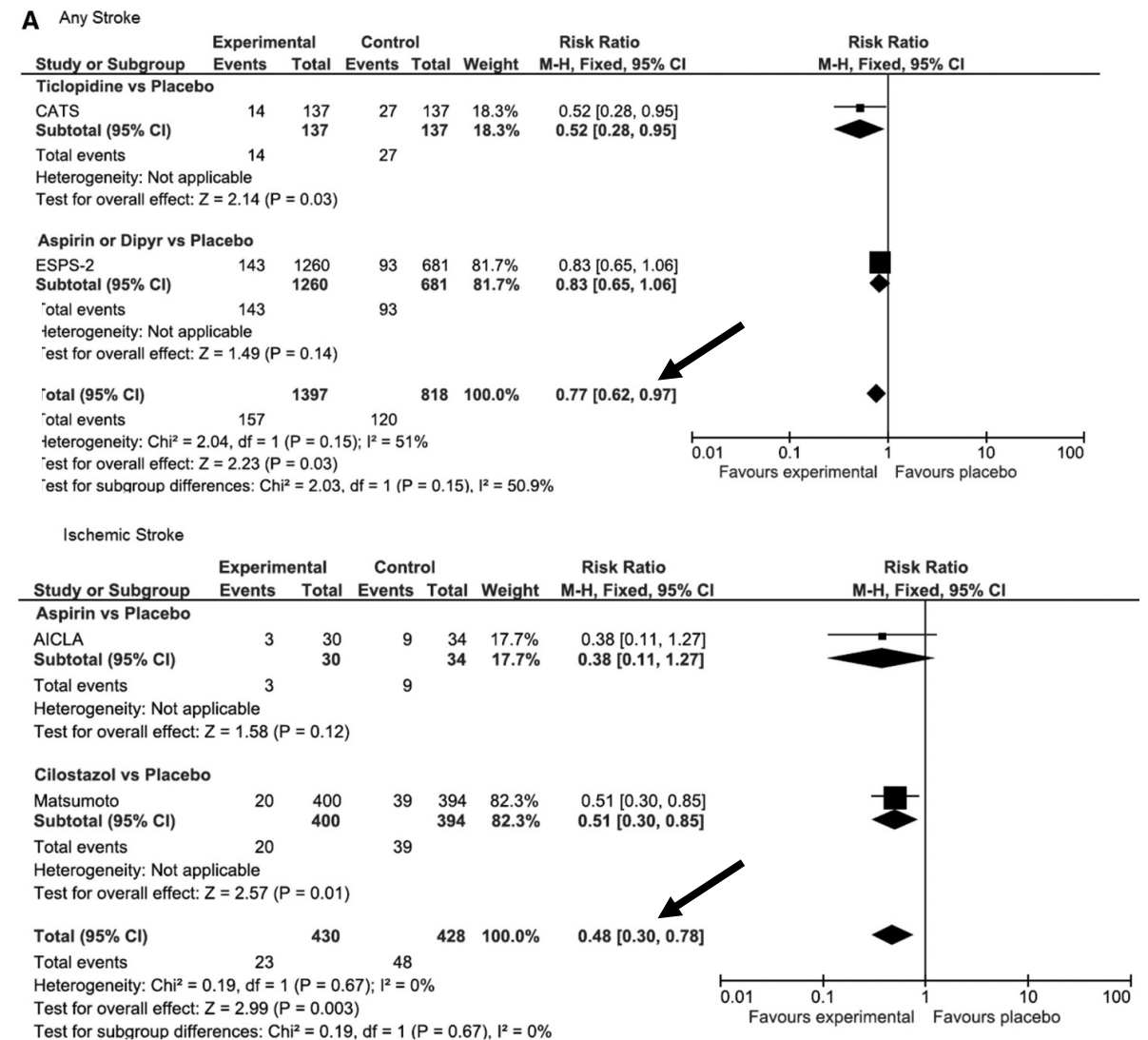
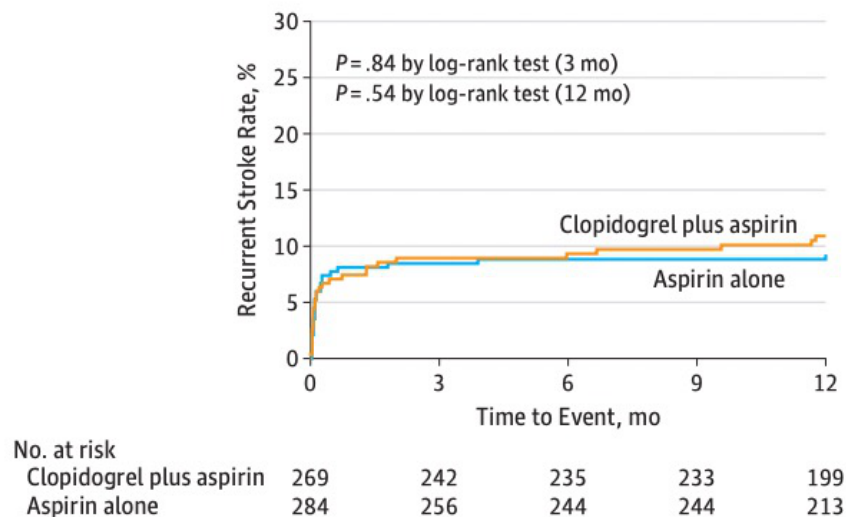
pubmed.ncbi.nlm.nih.gov/22463492/

pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

Secondary Prevention

- Antiplatelet Tx

C Single acute infarction



Stroke. 2015 Apr;46(4):1014-23.

JAMA Neurol. 2018 Jun 1;75(6):711-719.

Source-qualified teaching update | 30 September 2026

Short-course DAPT applies to selected acute noncardioembolic minor stroke/high-risk TIA, with initiation, duration and reperfusion exclusions specified by current guidance. The CHANCE single-acute-infarction subgroup is not synonymous with lacunar disease; the adjacent monotherapy-versus-placebo meta-analyses do not test DAPT versus monotherapy.

doi.org/10.1161/STR.0000000000000375

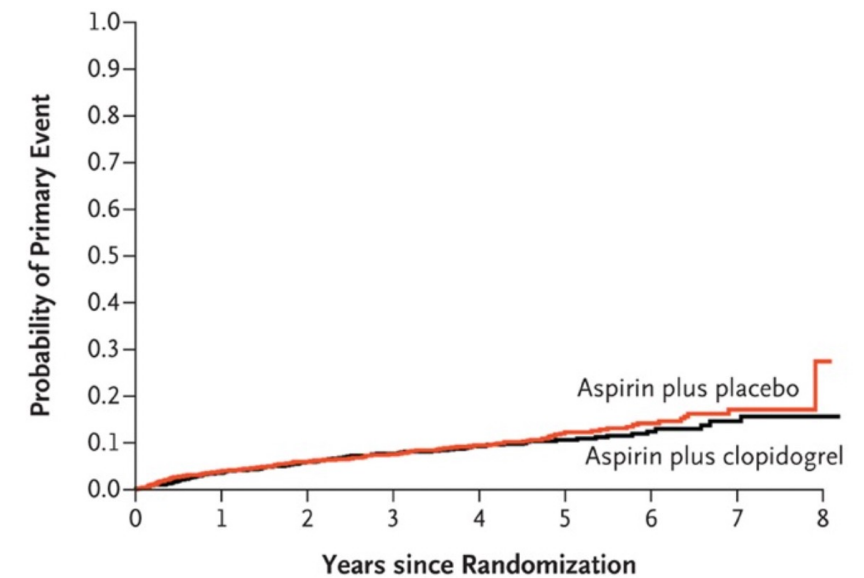
pmc.ncbi.nlm.nih.gov/articles/PMC10916806/

pmc.ncbi.nlm.nih.gov/articles/PMC5885215/

SPS3 (Secondary Prevention of Small Subcortical Strokes)

- P: recent symptomatic lacunar infarction (MRI)
 - I/C: 2x2 factorial design
 - **DAPT (ASA + clopidogrel) or ASA**
 - SBP<130 mmHg or SBP 130-149 mmHg
 - O: any recurrent stroke
- Recurrent stroke: HR 0.92 (95% CI 0.72 – 1.16)
- ↑ hemorrhage & mortality (HR 1.52)

N Engl J Med. 2012 Aug 30;367(9):817-25.



No. at Risk									
Aspirin plus placebo	1517	1272	1027	788	574	355	189	83	3
Aspirin plus clopidogrel	1503	1288	1030	802	589	371	205	90	5

Table 3. Safety Outcomes.*						
Outcome	Aspirin plus Placebo (N=1503)		Aspirin plus Clopidogrel (N=1517)		Hazard Ratio (95% CI)	P Value
	no.	rate (%/yr)	no.	rate (%/yr)		
All major hemorrhages	56	1.1	105	2.1	1.97 (1.41–2.71)	<0.001
Intracranial hemorrhages†	15*	0.28	22	0.42	1.52 (0.79–2.93)	0.21
Intracerebral	8	0.15	15	0.28	1.92 (0.82–4.54)	0.14
Subdural or epidural	6	0.11	7	0.13	1.23 (0.41–3.64)	0.72
Other	4	0.07	2	0.04	0.53 (0.10–2.89)	0.46
Extracranial bleeding	42	0.79	87	1.7	2.15 (1.49–3.11)	<0.001
Gastrointestinal‡	28	0.52	58	1.1	2.14 (1.36–3.36)	<0.001
Fatal hemorrhages	4	0.07	9	0.17	2.29 (0.70–7.42)	0.17
Intracranial	4	0.07	7	0.13	1.78 (0.52–6.07)	0.36
Extracranial	0	0	2	0.04	—	—

Source-qualified teaching update | 30 September 2026

Original-source inconsistency: SPS3 Figure 1 baseline risk-set labels show placebo 1517 and clopidogrel 1503, whereas Tables 1-3 report placebo 1503 and clopidogrel 1517. The figure is retained as a historical reproduction with this explicit caveat; no inference is made that later risk-set counts are also wrong.

SPS3 (Secondary Prevention of Small Subcortical Strokes)

- P: recent symptomatic lacunar infarction (MRI)
- I/C: 2x2 factorial design
 - DAPT (ASA + clopidogrel) or ASA
 - **SBP<130 mmHg or SBP 130-149 mmHg**
- O: any recurrent stroke

→ Recurrent stroke: HR 0.81 (95% CI 0.64 – 1.03)

- ↓ intracerebral hemorrhage

Lancet. 2013 Aug 10;382(9891):507-15.

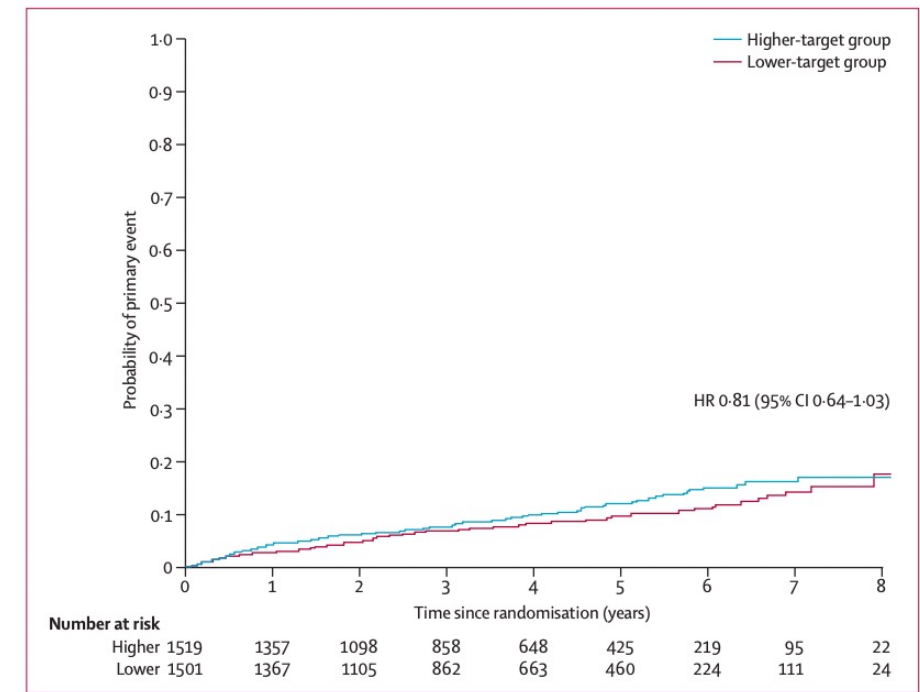
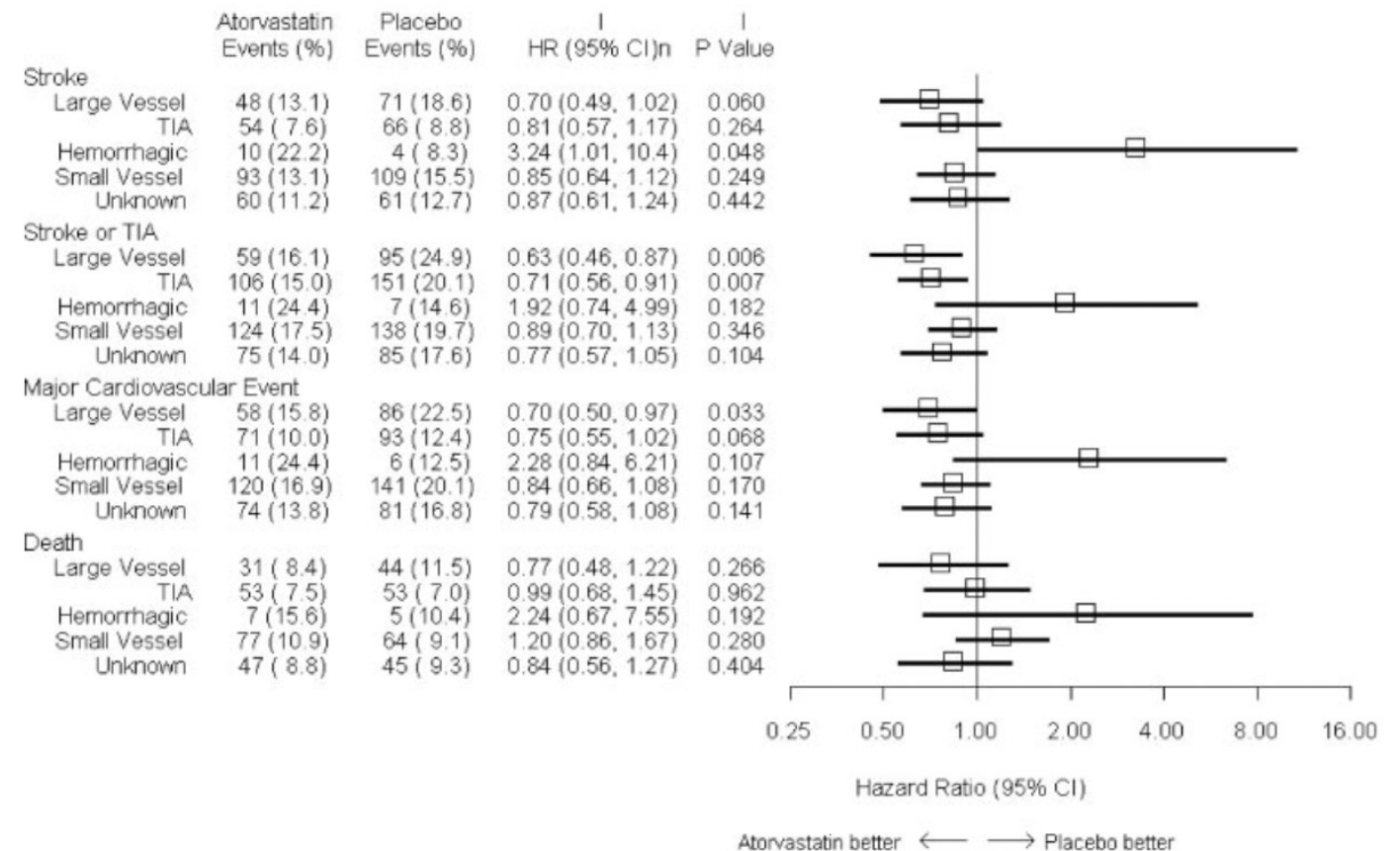


Figure 2: Probability of patients experiencing a primary event by time after randomisation

	Higher-target group (n=1519)		Lower-target group (n=1501)		Hazard ratio (95% CI)	p value
	Number of patients	Rate (% per patient-year)	Number of patients	Rate (% per patient-year)		
Stroke						
All stroke	152	2.77%	125	2.25%	0.81 (0.64-1.03)	0.08
Ischaemic stroke or unknown	131	2.4%	112	2.0%	0.84 (0.66-1.09)	0.19
Intracranial haemorrhage						
All	21*	0.38%	13†	0.23%	0.61 (0.31-1.22)	0.16
Intracerebral	16	0.29%	6	0.11%	0.37 (0.15-0.95)	0.03
Subdural or epidural	5	0.091%	6	0.11%	1.18 (0.36-3.88)	0.78
Other	2	0.036%	4	0.072%	1.97 (0.36-10.74)	0.43
Disabling or fatal stroke‡	49	0.89%	40	0.72%	0.81 (0.53-1.23)	0.32

Secondary Prevention

- Blood Pressure
 - Long-term target SBP<130 mmHg
- Lifestyle modifications
 - Tobacco cessation, Limit ETOH
 - Physical activity
 - Mediterranean diet (HTN - Na restriction)



Stroke. 2009 Apr;40(4):1405-9.

Source-qualified teaching update | 30 September 2026

Lipid treatment should follow the patient's atherosclerotic/ASCVD risk and current guidance. The 2021 LDL-C <70 mg/dL recommendation specifically concerns stroke/TIA with atherosclerotic disease; SPARCL tested atorvastatin 80 mg, not randomized LDL targets. This is not a reason to withhold indicated statin treatment.

doi.org/10.1161/STR.0000000000000375
pmc.ncbi.nlm.nih.gov/articles/PMC10916806/
pubmed.ncbi.nlm.nih.gov/19228842/

Concurrent Potential Causes

Carotid Stenosis

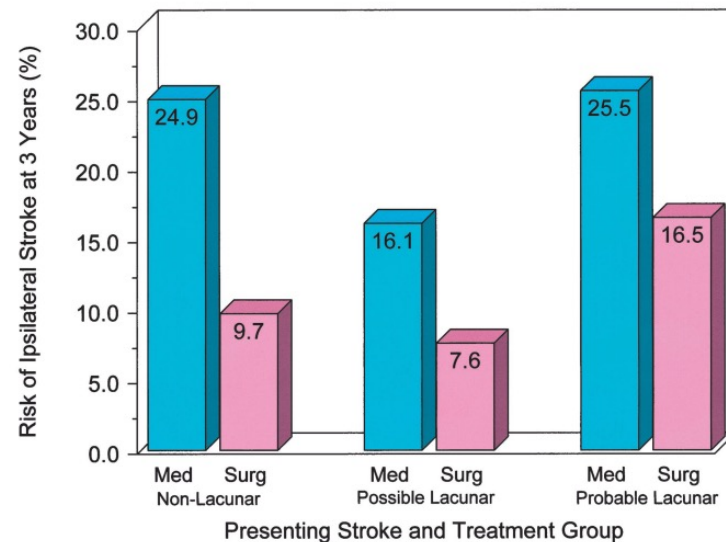


Figure 4. Risk of ipsilateral stroke at 3 years by presenting stroke category and treatment group for patients with 50 to 99% internal carotid artery stenosis. The numbers of patients represented in each bar, from left to right, are 172, 160, 57, 69, 41, and 38. Med = medically treated; Surg = surgically treated.

Atrial Fibrillation

Figure 2. Time to First Detection of Atrial Fibrillation at 12 Months in a Study of Long-term Continuous Cardiac Monitoring vs Usual Care on Detection of Atrial Fibrillation (AF) in Patients With Stroke Attributed to Large- or Small-Vessel Disease

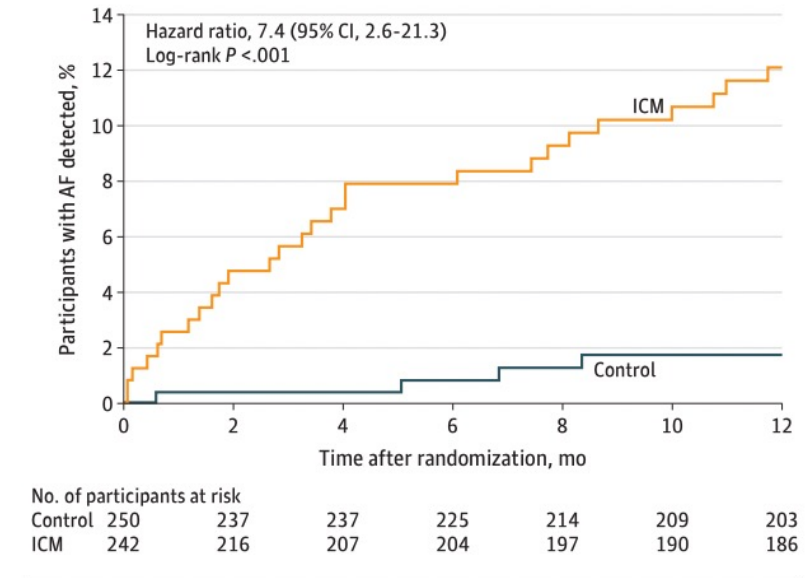
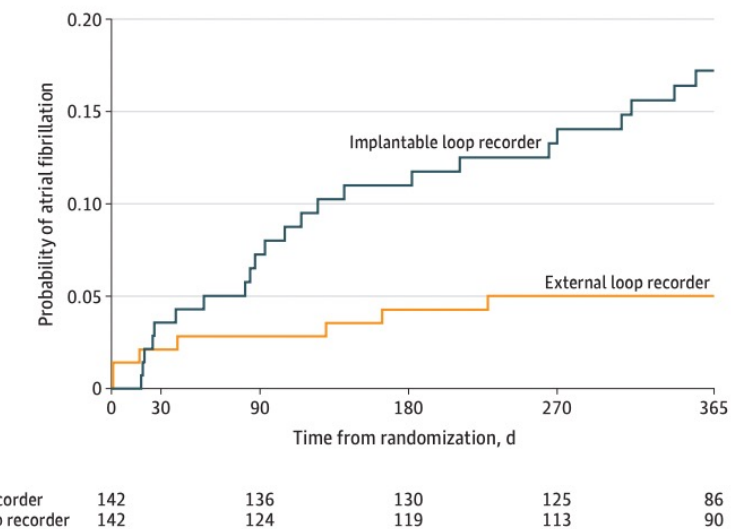


Figure 2. Secondary Outcome of the Probability of New Atrial Fibrillation or Flutter Lasting 2 Minutes or Longer Within 12 Months



Prognosis

- Early Neurological Deterioration
 - IL-6, TNF- α , ICAM-1
 - Preceding TIA, fluctuating course
- Functional Recovery
 - Lower infarct volumes
 - Late recovery

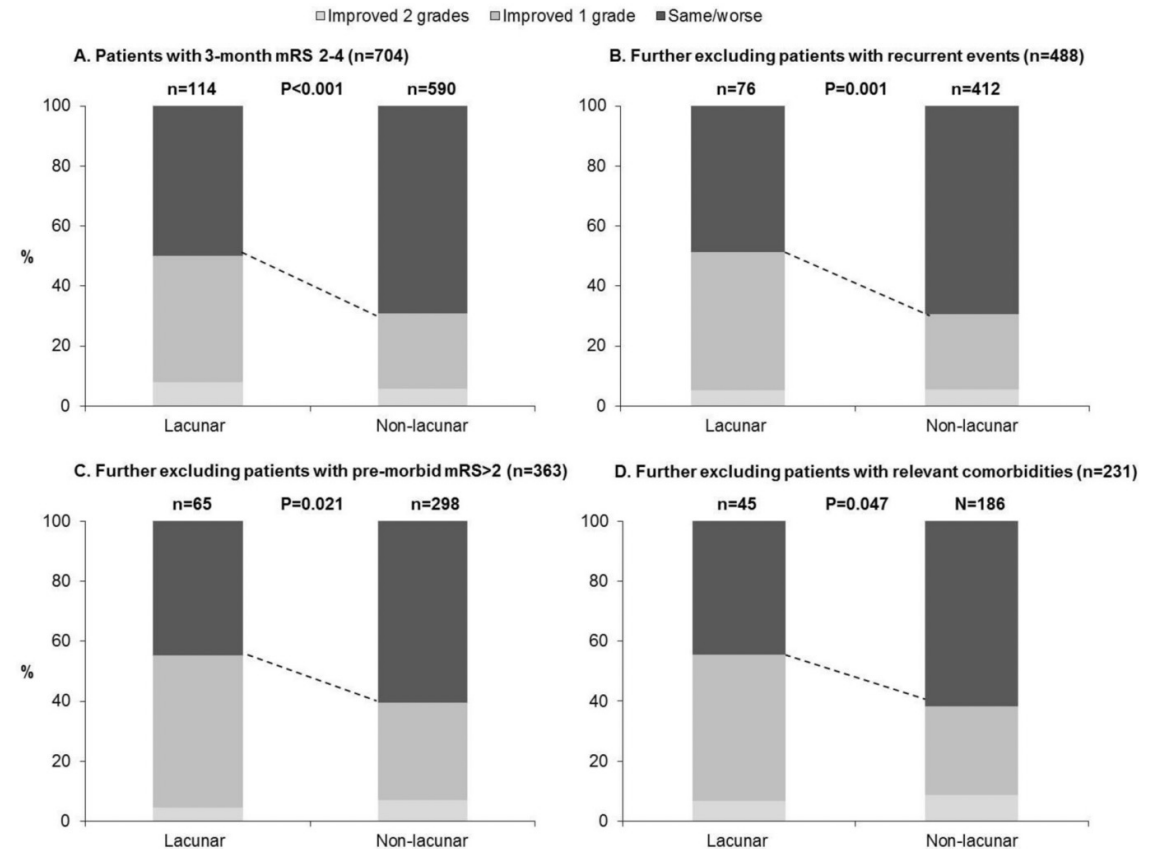
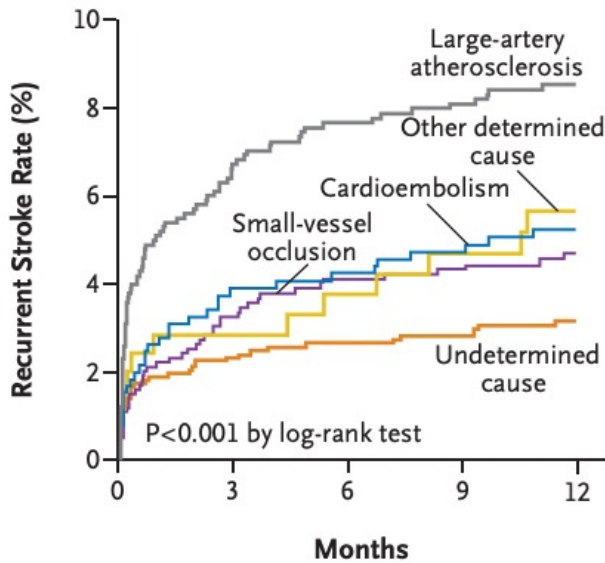


Figure 2 Changes in mRS between 3-months and 1-year post-stroke for 3-month survivors of lacunar versus non-lacunar stroke, (A) including all patients with 3-month mRS of 2 to 4, and then progressively excluding patients with: (B) recurrent vascular events over follow-up, (C) pre-morbid mRS>2, and (D) relevant comorbidities including peripheral vascular disease, heart failure, valve disease, and/or cancer. P-values are from Wilcoxon rank-sum tests for trend.

J Neurol Neurosurg Psychiatry. 2018 Dec;89(12):1301-1307.

Recurrent Stroke & Survival

D Rate of Recurrent Stroke According to Cause of TIA or Minor Stroke (TOAST Classification)



No. at Risk					
	0	3	6	9	12
Large-artery atherosclerosis	987	892	863	853	799
Small-vessel occlusion	983	905	862	857	790
Cardioembolism	641	584	570	561	494
Other determined cause	244	214	205	198	184
Undetermined cause	1354	1263	1206	1199	1085

Brain. 2005 Nov;128(Pt 11):2507-17.

N Engl J Med. 2016 Apr 21;374(16):1533-42.

N Engl J Med. 2018 Jun 7;378(23):2182-2190.

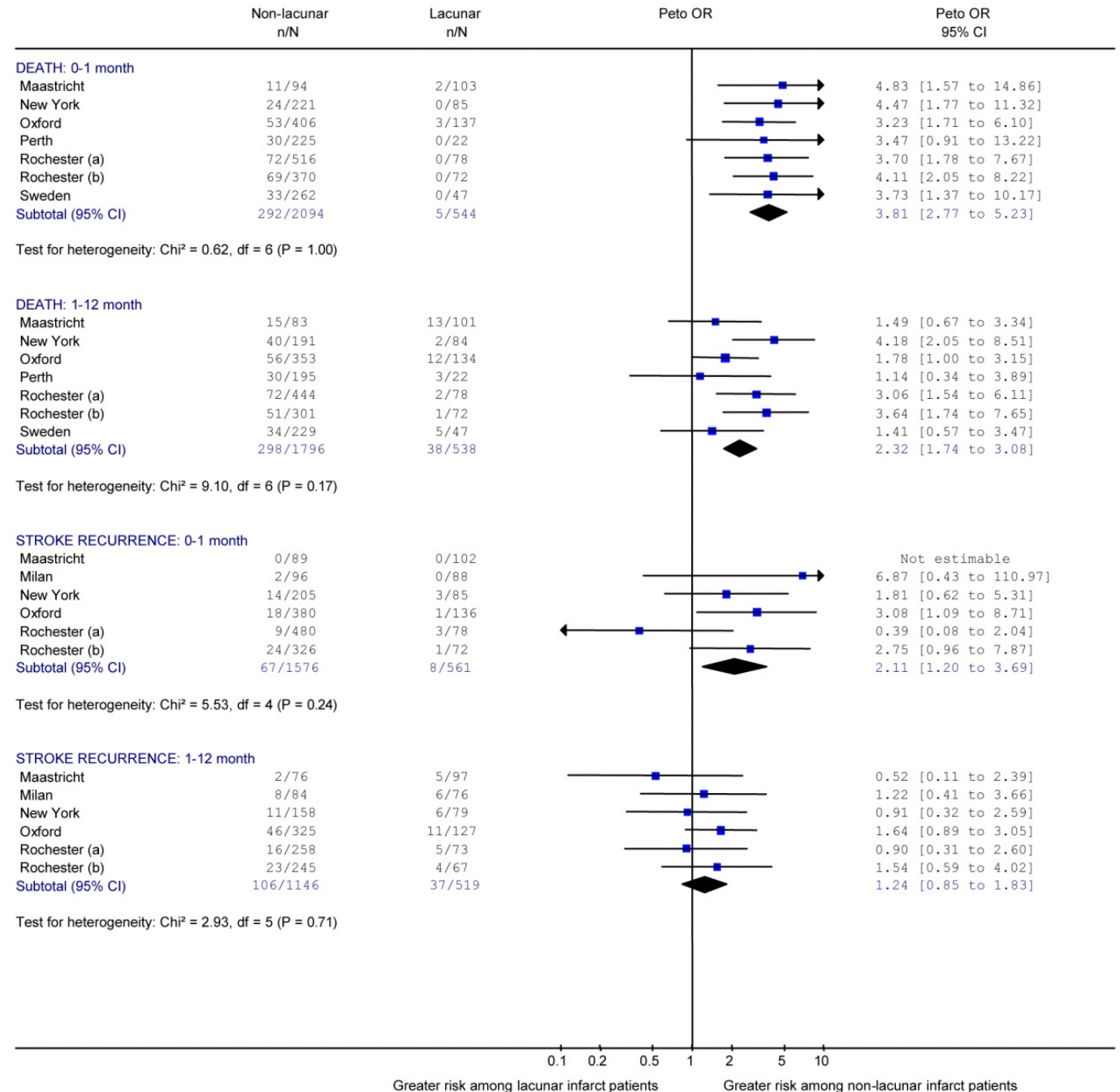
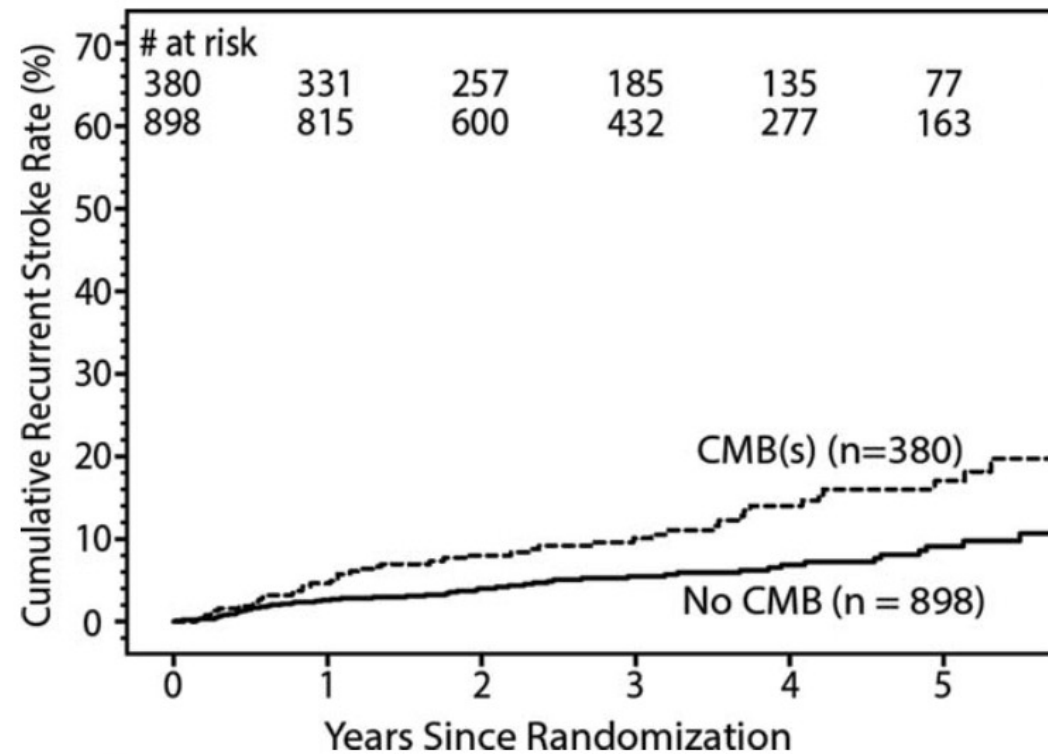


Table 2. *Factors associated with recurrent ischemic stroke: multivariable analysis*

Feature	Derivation cohort hazard ratio (95% CI)	Validation cohort hazard ratio (95% CI)	All participants hazard ratio (95% CI)
Male sex	1.7 (1.1, 2.6)	1.3 (.9, 1.9)	1.5 (1.1, 1.9)
Black race	2.0 (1.4, 3.0)	1.5 (1.0, 2.3)	1.7 (1.3, 2.3)
Diabetes	2.1 (1.5, 3.0)	1.9 (1.3, 2.7)	2.0 (1.5, 2.5)
Prior symptomatic lacunar stroke or TIA	2.5 (1.7, 3.7)	1.9 (1.3, 2.8)	2.2 (1.6, 2.9)

**FIGURE 1:** Kaplan–Meier curve for recurrent stroke by cerebral microbleed (CMB) status.

Uncertainties and Opportunities

- Management
 - Microangiopathy
 - Branch atheromatous disease
 - Novel therapeutics
- Epidemiology
- VCID
- Subclinical infarcts
 - $\geq 5X$ more common
 - Majority lacunar

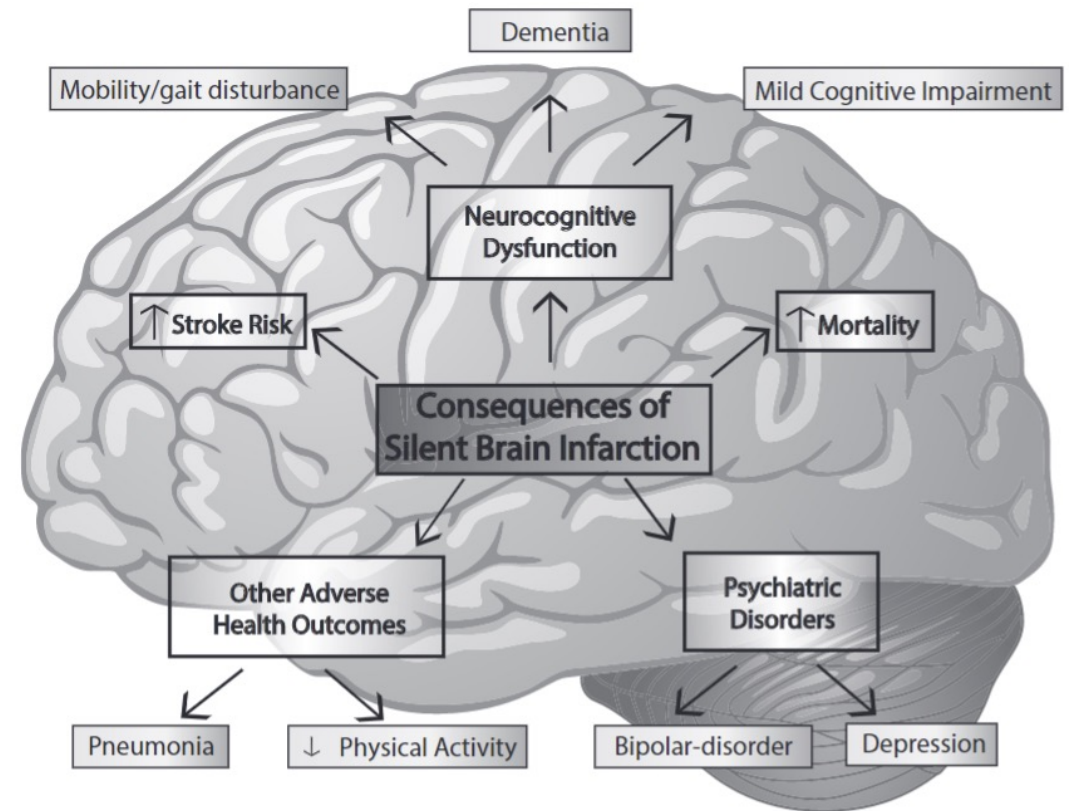


Figure 5. Consequences of silent brain infarction.

Questions?